

Erste Schritte im IoT mit LoRaWAN

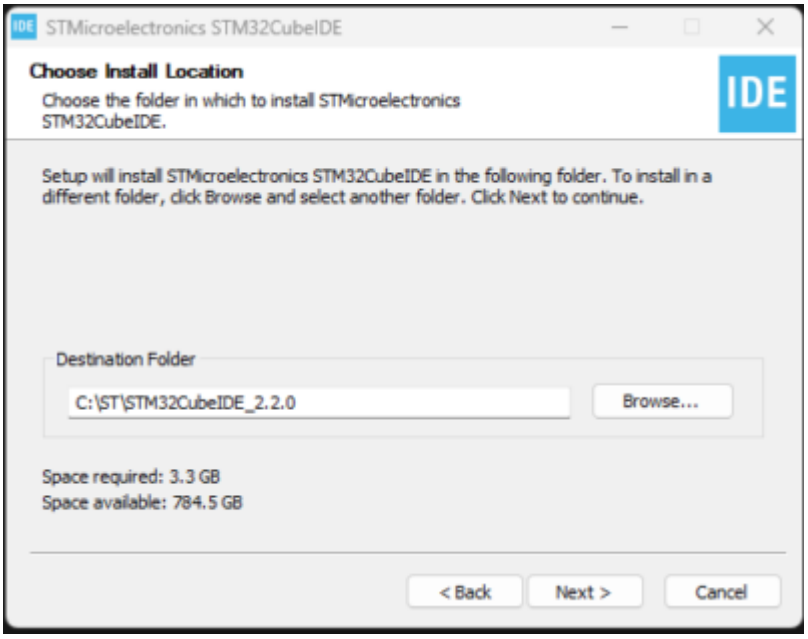
Entwicklungsumgebung

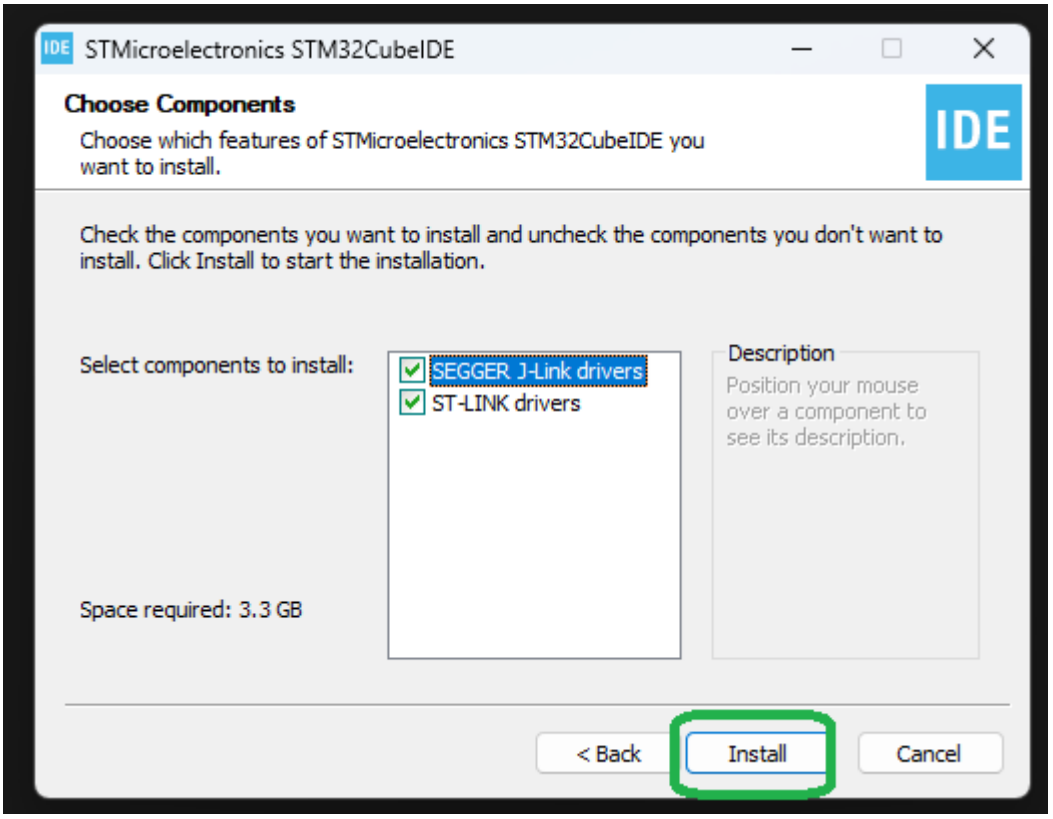
- Hardware/Firmware
    - Gateway(optional): NUCLEO-F746ZG + LRWAN GS HF1 LoRa
    - Entwicklungsboard: STM32WL55JC1, STM32WL55JCIx Firmware: V1.3.1 + Treiber
    - Apiariot-PCB: Demnächst verfügbar.
- *Software:*
    - STMCubeIDE (V:2.2.0)
    - STM32CubeMX (V:6.18.1)
    - STMCubeWL (V1.6.0/1): Bibliothek für STM32WL55JC
    - Entwicklungsboard STM32WL55JCIx, Firmware: V1.3.

Vorbereitung der Umgebung (Toolchain)

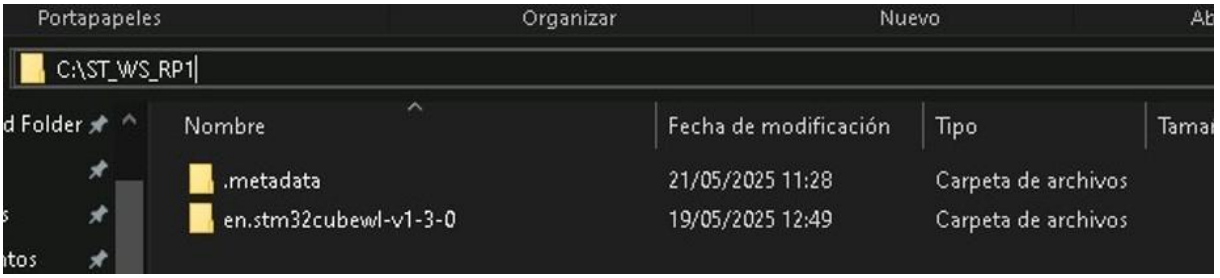
Es wird empfohlen, denselben Ordnerpfad beizubehalten und jeden Schritt genau zu befolgen.

- Lade die integrierte Entwicklungsumgebung (IDE) “STM32CubeIDE” herunter und installiere sie.

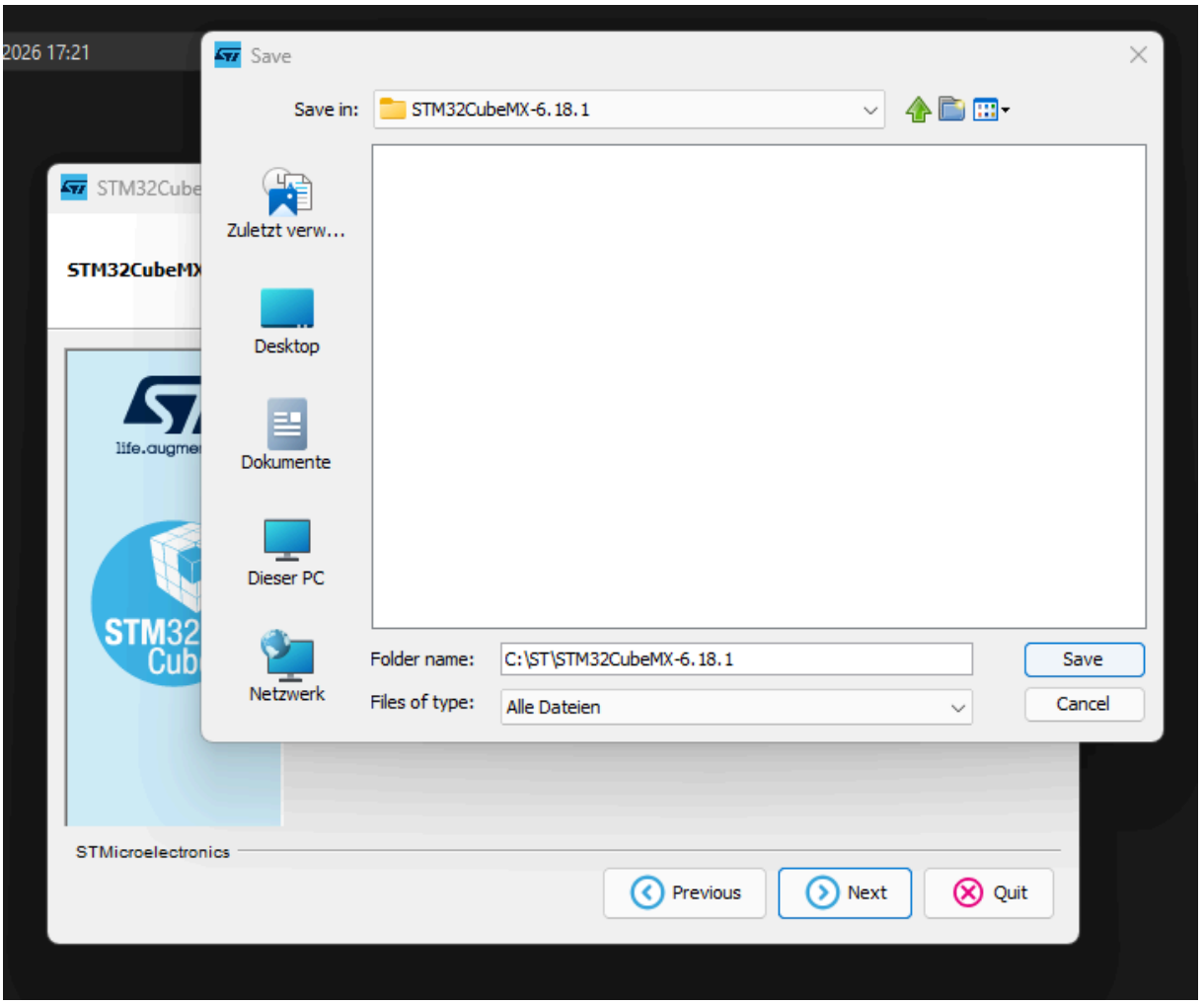


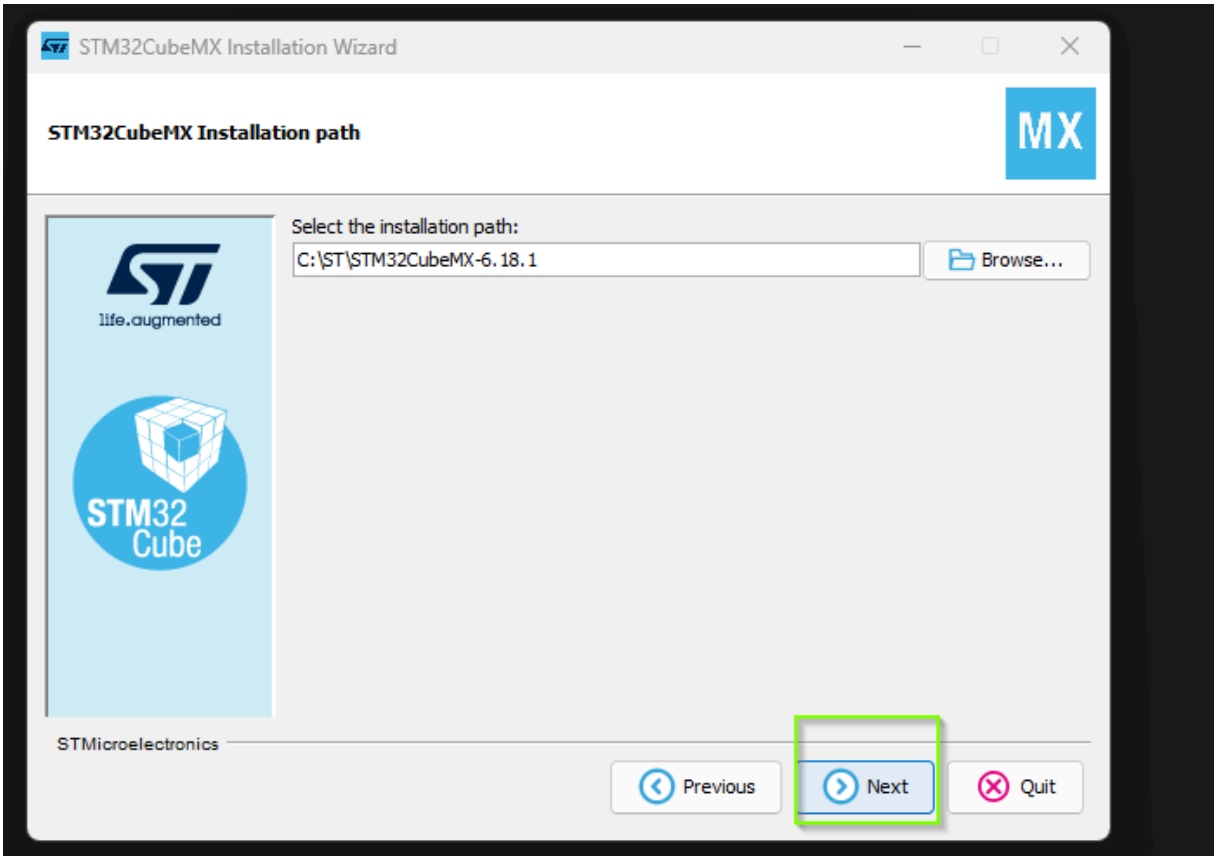


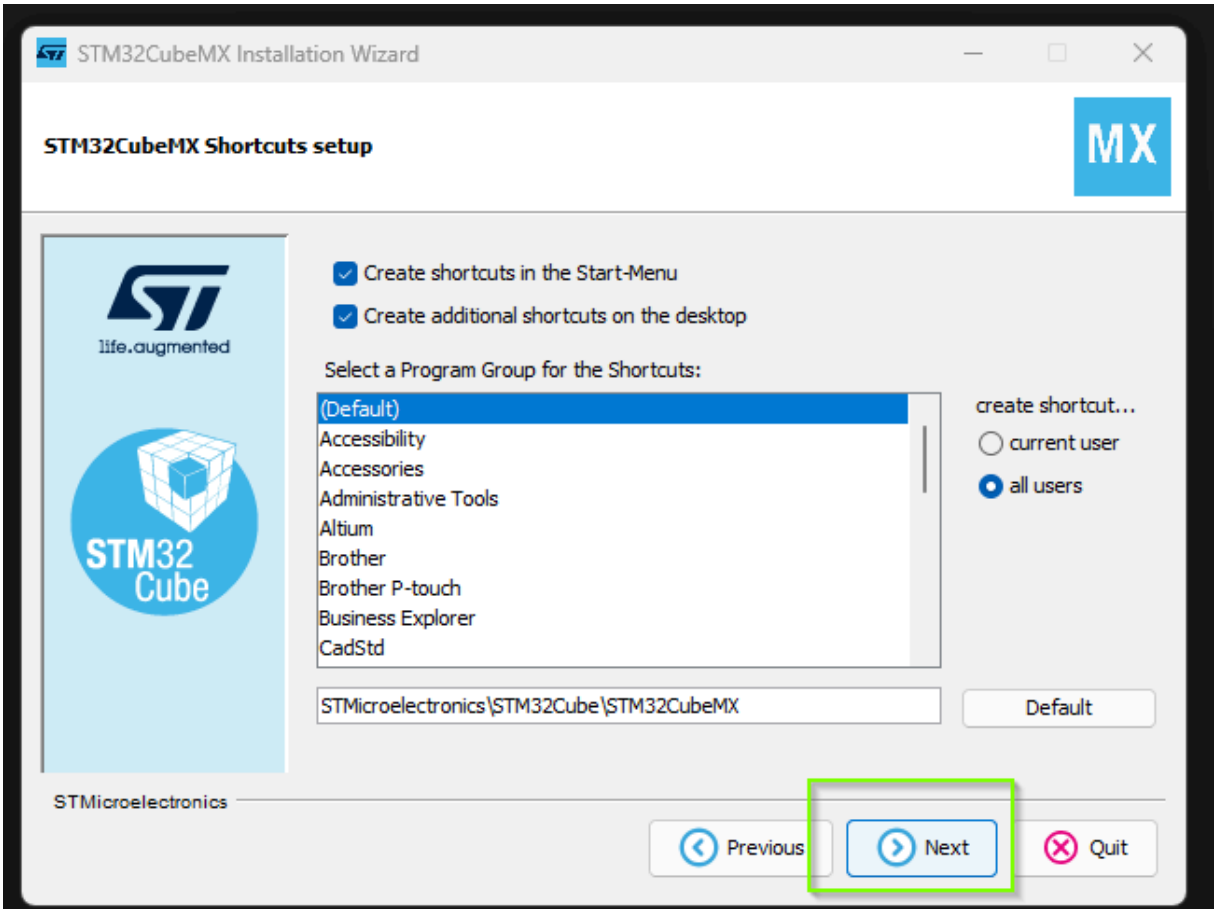
- Erstelle den Ordner "C:ST_WS_RP1".
  Es wird empfohlen, denselben Ordner und denselben Pfad zu verwenden.



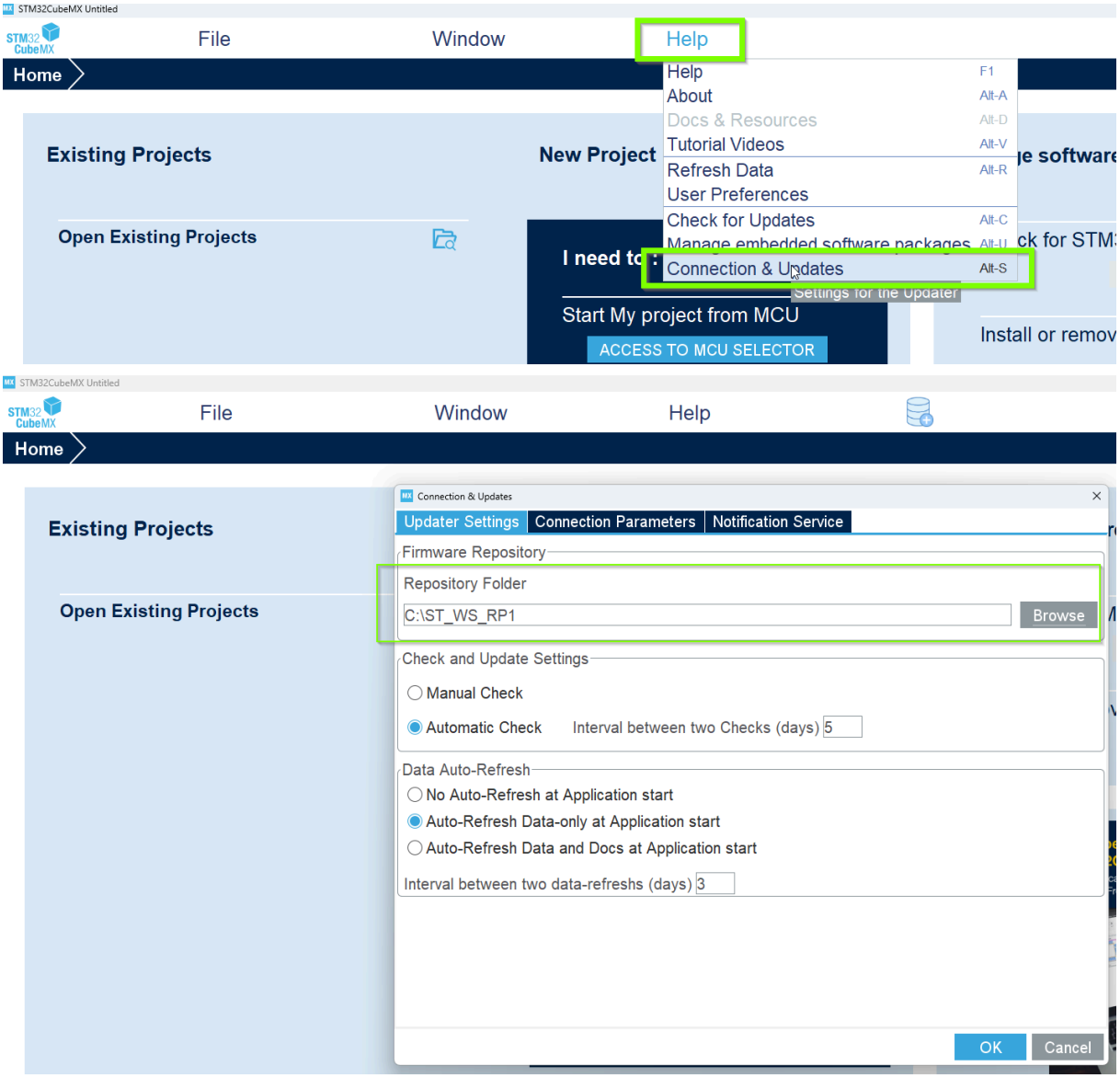
- Erstelle den Ordner "C:ST\STM32CubeMX-6.18.1 ", lade STM32CubeMX (V:6.18.1) herunter und installiere es.

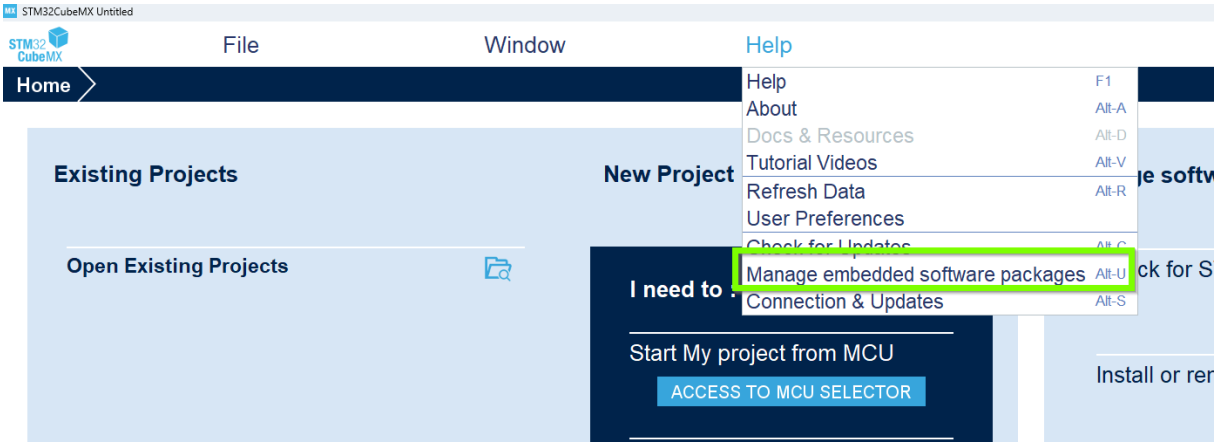


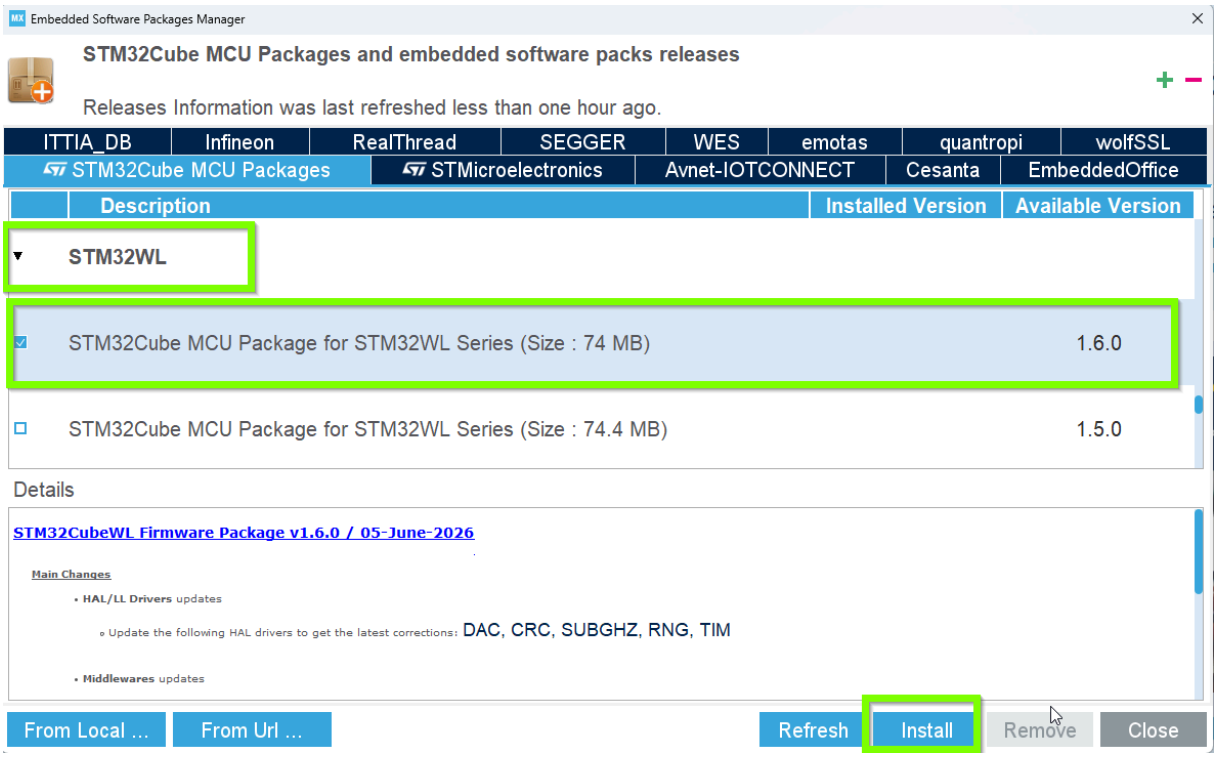

STM32CubeMX Installation Wizard
STM32CubeMX Installation path
MX
life.augmented
STM32 Cube
Select the installation path:
C:\ST\STM32CubeMX-6.18.1
Browse...
STMicroelectronics
Previous
Next
Quit


STM32CubeMX Installation Wizard
STM32CubeMX Shortcuts setup
MX
life.augmented
STM32 Cube
Create shortcuts in the Start-Menu
Create additional shortcuts on the desktop
Select a Program Group for the Shortcuts:
(Default)
Accessibility
Accessories
Administrative Tools
Altium
Brother
Brother P-touch
Business Explorer
CadStd
create shortcut...
current user
all users
STMicroelectronics\STM32Cube\STM32CubeMX
Default
STMicroelectronics
Previous
Next
Quit

- Starte STM32CubeMX



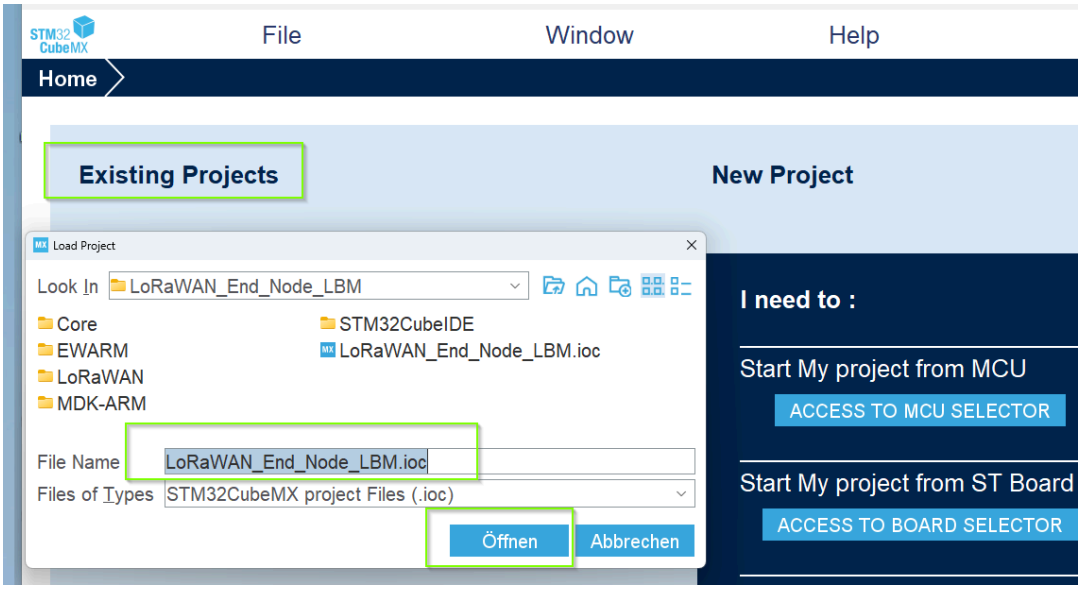

STM32CubeMX Untitled
File
Window
Help
Home
Help F1
About Alt-A
Docs & Resources Alt-D
Tutorial Videos Alt-V
Refresh Data Alt-R
User Preferences
Manage embedded software packages Alt-U
Connection & Updates Alt-S
Existing Projects
New Project
Open Existing Projects
I need to :
Start My project from MCU
ACCESS TO MCU SELECTOR
Install or rer

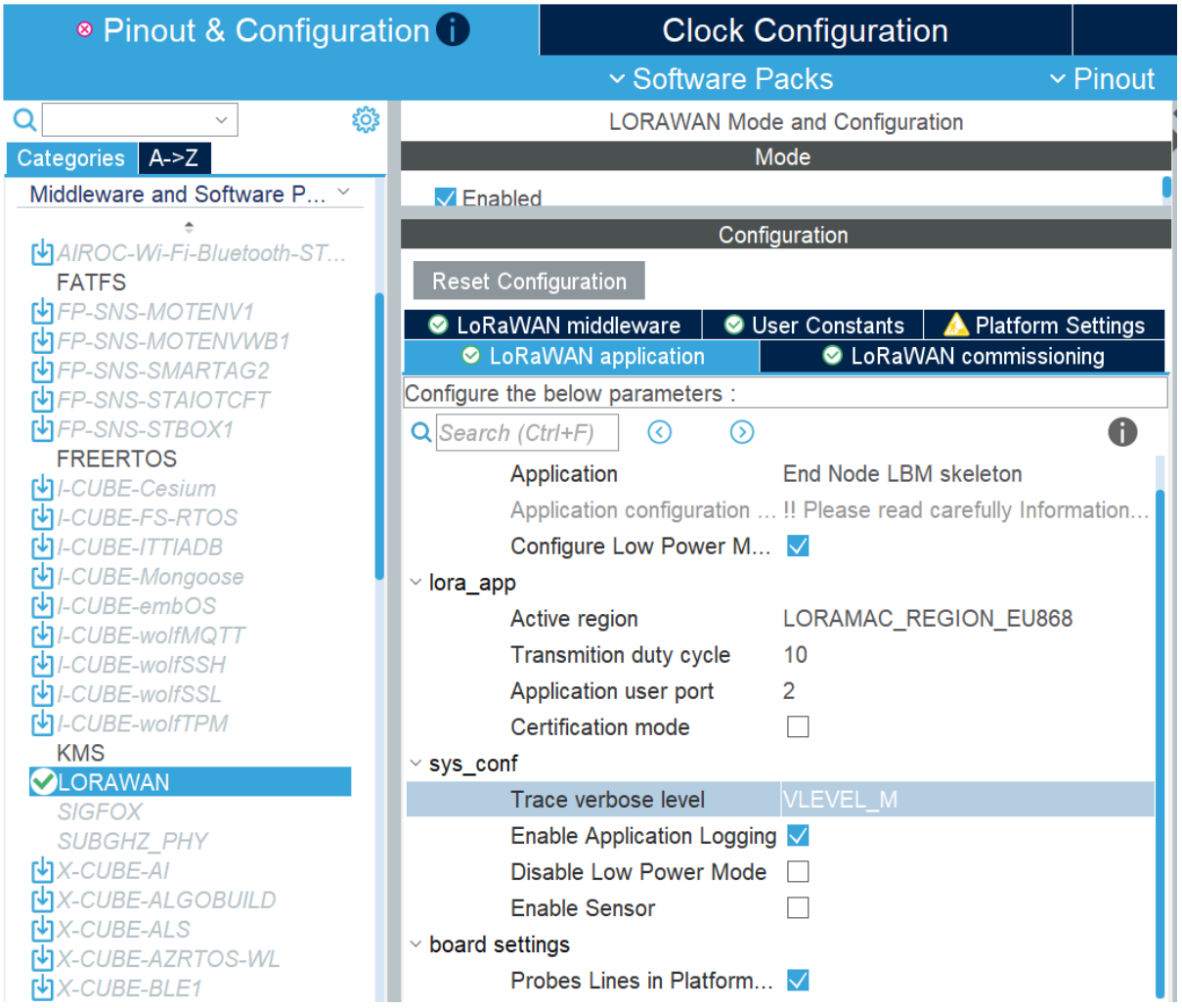

Embedded Software Packages Manager
STM32Cube MCU Packages and embedded software packs releases
Releases Information was last refreshed less than one hour ago.
ITTIA_DB
Infineon
RealThread
SEGGER
WES
emotas
quantropi
wolfSSL
STM32Cube MCU Packages
STMicroelectronics
Avnet-IOTCONNECT
Cesanta
EmbeddedOffice
Description
Installed Version
Available Version
STM32WL
STM32Cube MCU Package for STM32WL Series (Size : 74 MB) 1.6.0
STM32Cube MCU Package for STM32WL Series (Size : 74.4 MB) 1.5.0
Details
STM32CubeWL Firmware Package v1.6.0 / 05-June-2026
Main Changes
HAL/LL Drivers updates
Update the following HAL drivers to get the latest corrections: DAC, CRC, SUBGHZ, RNG, TIM
Middlewares updates
From Local ...
From Url ...
Refresh
Install
Remove
Close

LoRaWAN Konfiguration:

- Starte: LoRaWAN_End_Node_LBM.ioc

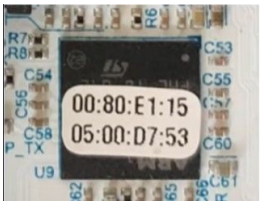
  C:\ST_WS_RP1\STM32Cube_FW_WL_V1.6.0\Projects\NUCLEO-WL55JC\Applications\LoRaWAN\LoRaWAN_End_Node_LBM\LoRaWAN_End_Node_LBM.ioc





LoRaWAN commissioning:

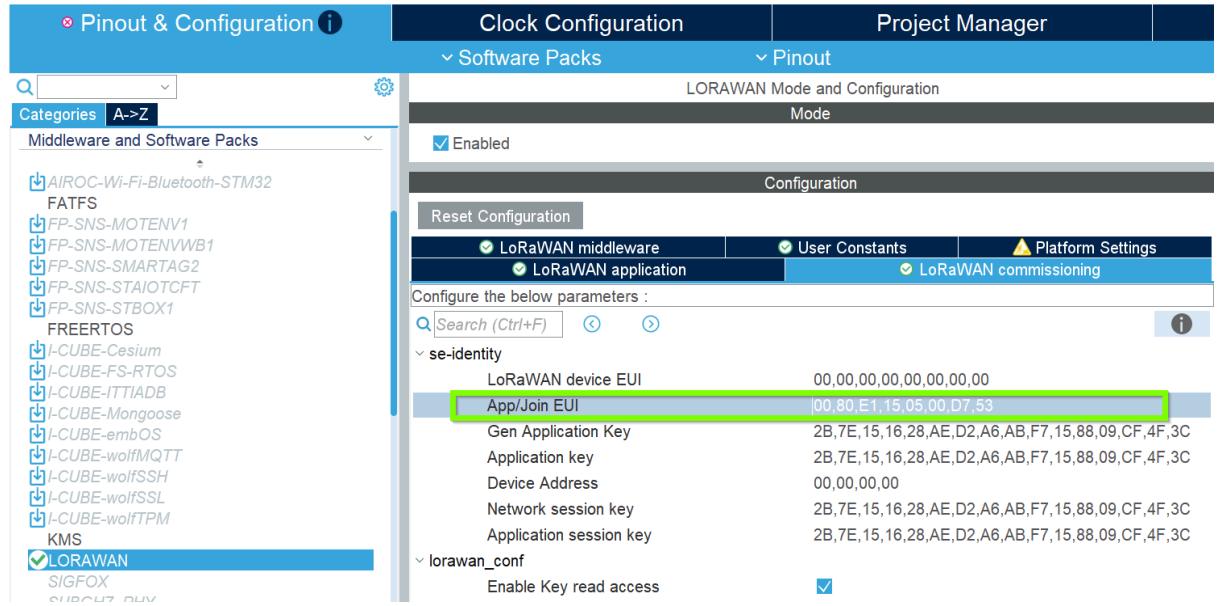
- *“App/Join EUI”*: Ist auf dem Board zu finden.

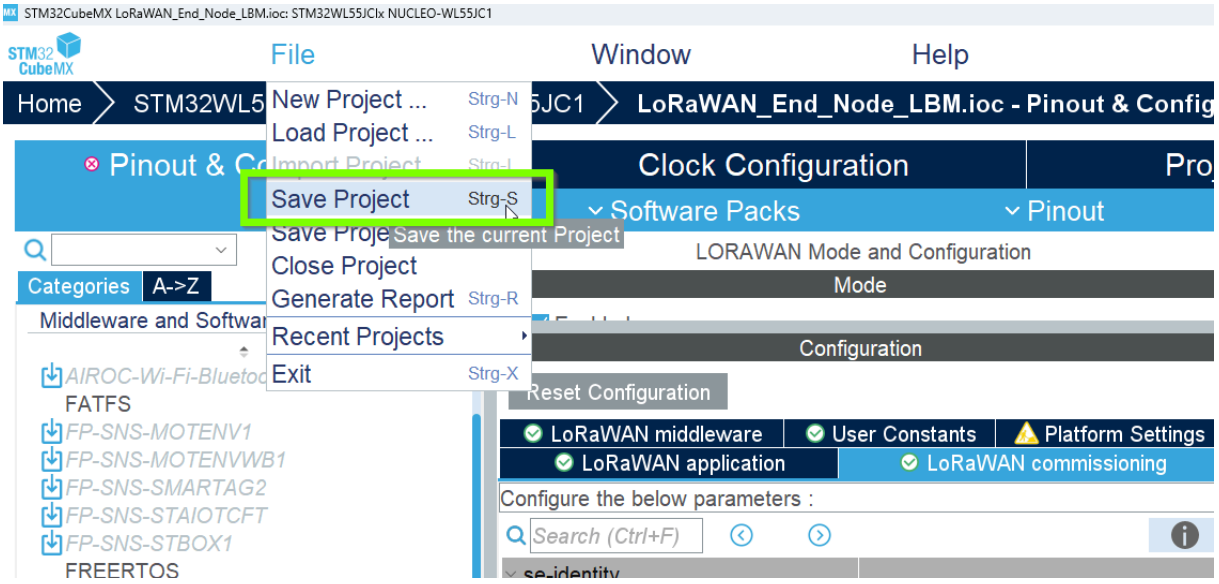


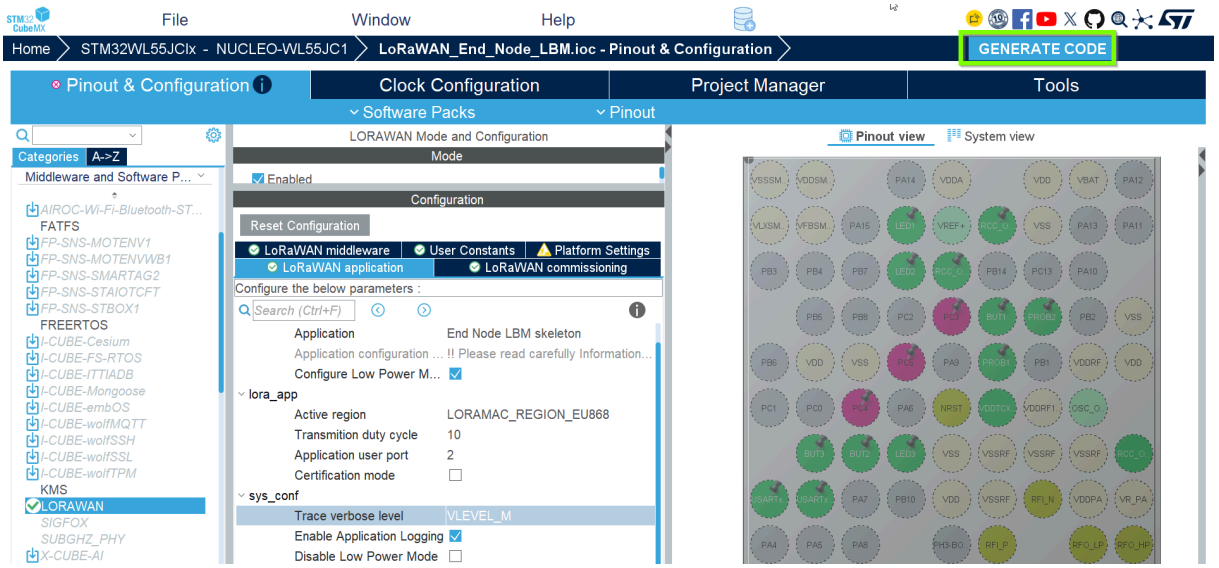
Wichtig:Das Feld "LORAWAN_device EUI" sollte NICHT als "00.00.00.00.00.00.00.00" definiert bleiben. Selbst wählen.

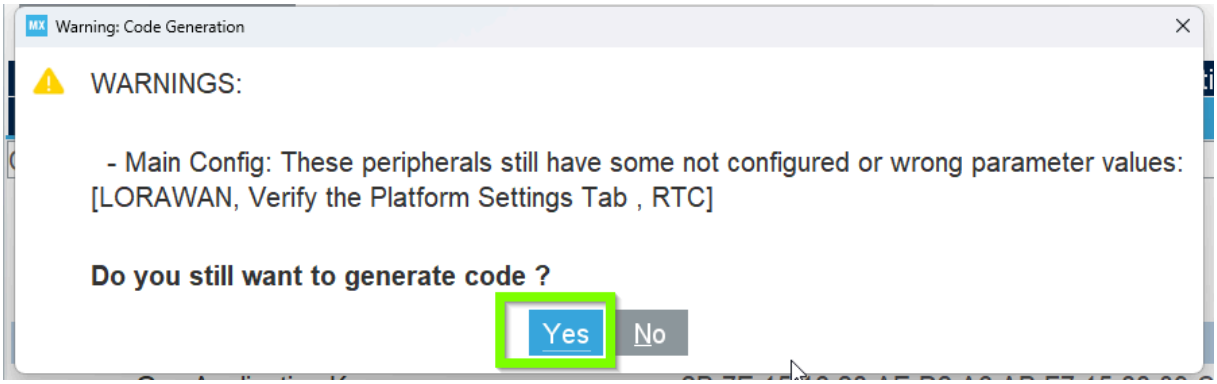
```
/*!
* End-device IEEE EUI (big endian)
* When set to 00,00,00,00,00,00,00,00 DevEui is automatically set
with a value provided by MCU platform
*/
#define LORAWAN_DEVICE_EUI
00,00,00,00,00,00,00,00
```

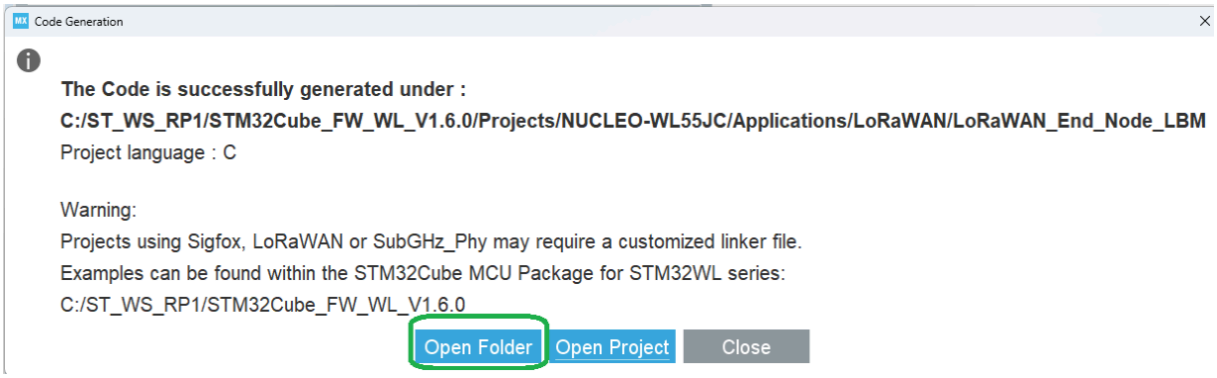
- *“Application key”: selbst wählen*


Pinout & Configuration
Clock Configuration
Project Manager
Software Packs
Pinout
Categories
A->Z
Middleware and Software Packs
AIROC-Wi-Fi-Bluetooth-STM32
FATFS
FP-SNS-MOTENV1
FP-SNS-MOTENVWB1
FP-SNS-SMARTAG2
FP-SNS-STAIOTCFT
FP-SNS-STBOX1
FREERTOS
I-CUBE-Cesium
I-CUBE-FS-RTOS
I-CUBE-ITTIADB
I-CUBE-Mongoose
I-CUBE-embOS
I-CUBE-wolfMQTT
I-CUBE-wolfSSH
I-CUBE-wolfSSL
I-CUBE-wolfTPM
KMS
LORAWAN
SIGFOX
LORAWAN Mode and Configuration
Mode
Enabled
Configuration
Reset Configuration
LoRaWAN middleware
User Constants
Platform Settings
LoRaWAN application
LoRaWAN commissioning
Configure the below parameters :
Search (Ctrl+F)
se-identity
LoRaWAN device EUI
00,00,00,00,00,00,00,00
App/Join EUI
00,80,E1,15,05,00,D7,53
Gen Application Key
2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,09,CF,4F,3C
Application key
2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,09,CF,4F,3C
Device Address
00,00,00,00
Network session key
2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,09,CF,4F,3C
Application session key
2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,09,CF,4F,3C
lorawan_conf
Enable Key read access

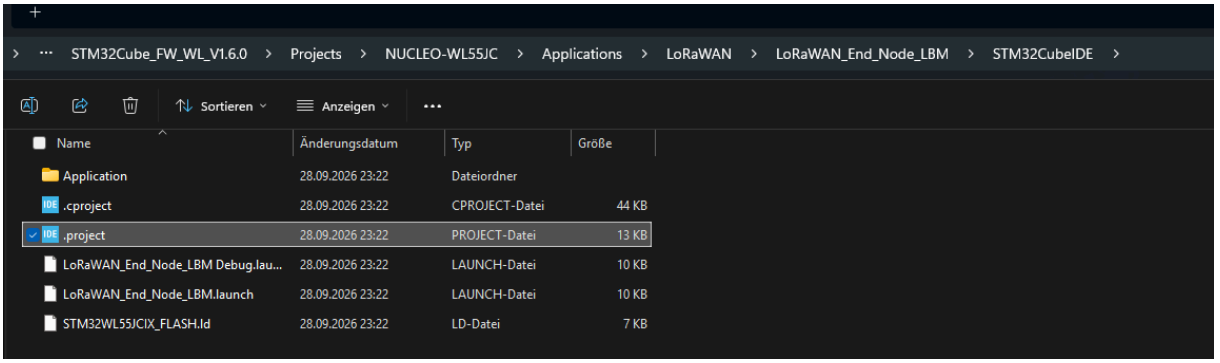

STM32WL55JCIx [NUCLEO-WL55JC1]
File
Window
Help
Home
LoRaWAN_End_Node_LBM.ioc - Pinout & Config
New Project ... Strg-N
Load Project ... Strg-L
Save Project Strg-S
Save the current Project
Close Project
Generate Report Strg-R
Recent Projects
Exit Strg-X
Clock Configuration
Software Packs
Pinout
LORAWAN Mode and Configuration
Mode
Configuration
Reset Configuration
LoRaWAN middleware
User Constants
Platform Settings
LoRaWAN application
LoRaWAN commissioning
Configure the below parameters :
Search (Ctrl+F)


File
Window
Help
Home
STM32WL55JCIx - NUCLEO-WL55JC1
LoRaWAN_End_Node_LBM.ioc - Pinout & Configuration
GENERATE CODE
Pinout & Configuration
Clock Configuration
Project Manager
Tools
Software Packs
Pinout
LORAWAN Mode and Configuration
Mode
Enabled
Configuration
Reset Configuration
LoRaWAN middleware
User Constants
Platform Settings
LoRaWAN application
LoRaWAN commissioning
Configure the below parameters :
Search (Ctrl+F)
Application
End Node LBM skeleton
Application configuration ...
!! Please read carefully Information...
Configure Low Power M...
lora_app
Active region
LORAMAC_REGION_EU868
Transmition duty cycle
10
Application user port
2
Certification mode
sys_conf
Trace verbose level
VLEVEL_M
Enable Application Logging
Disable Low Power Mode
Pinout view
System view

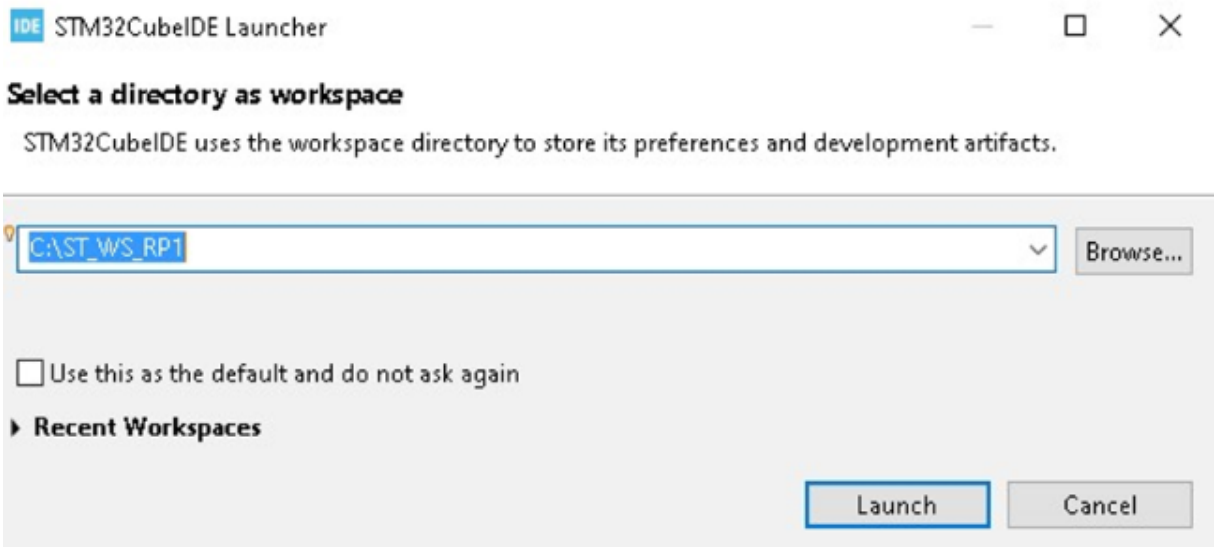


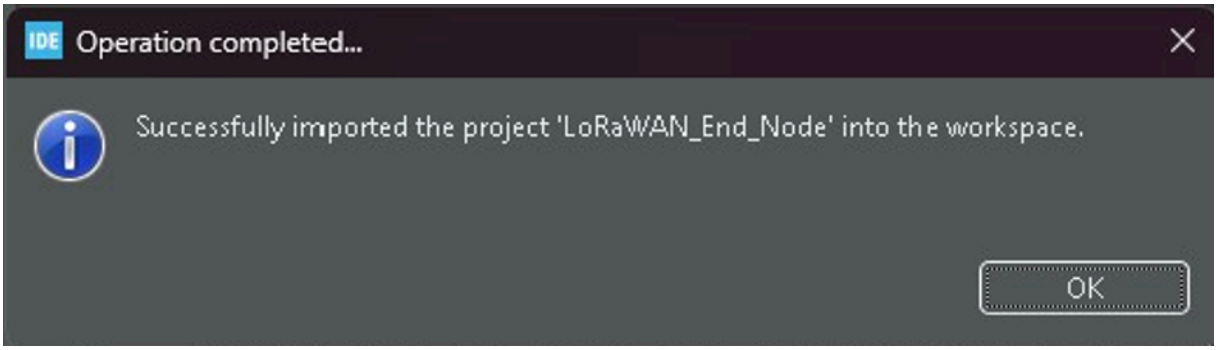


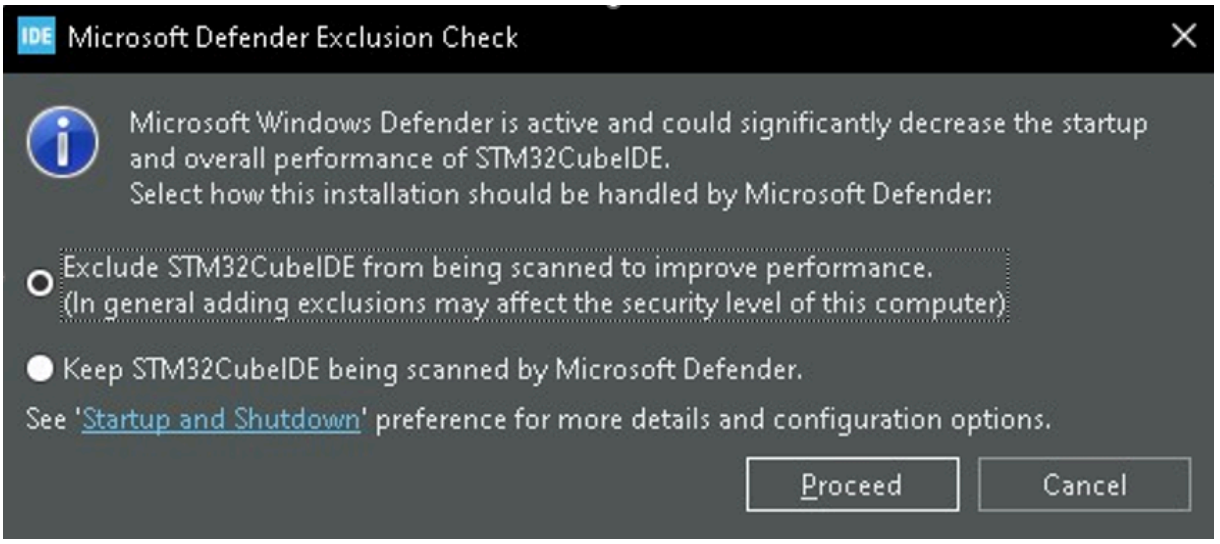
- Starte das Projekt mit STM32cubeIDE
  "C:\ST_WS_RP1\STM32Cube_FW_WL_V1.6.0\Projects\NUCLEO-WL55JC\Applications\LoRaWAN\LoRaWAN_End_Node_LBM\STM32CubeIDE\.project"

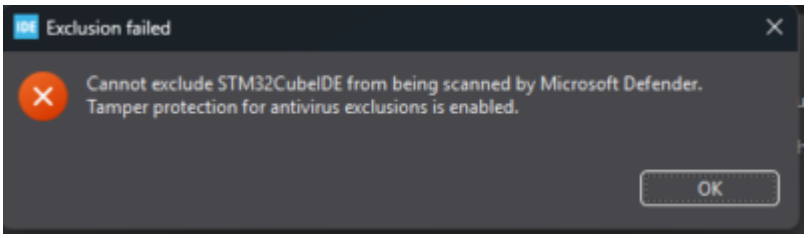


- Wähle als Workspace C:ST_WS_RP1

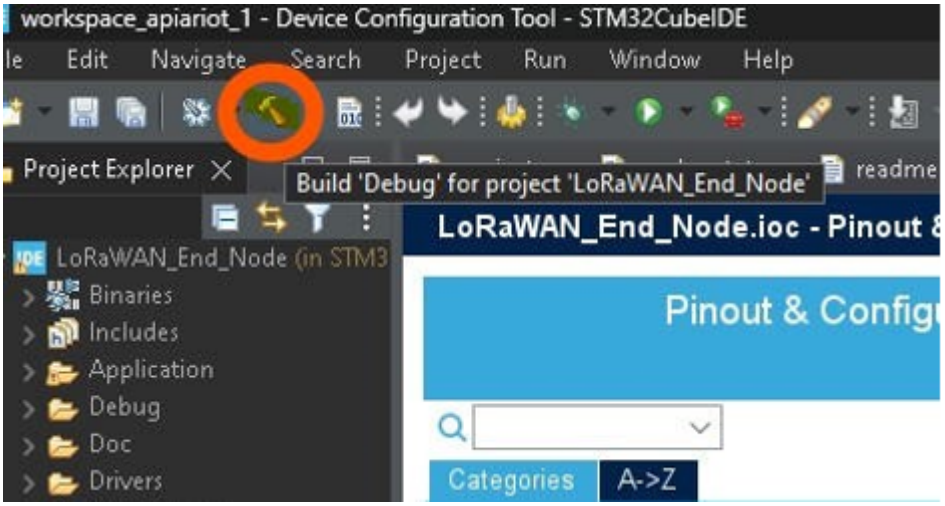


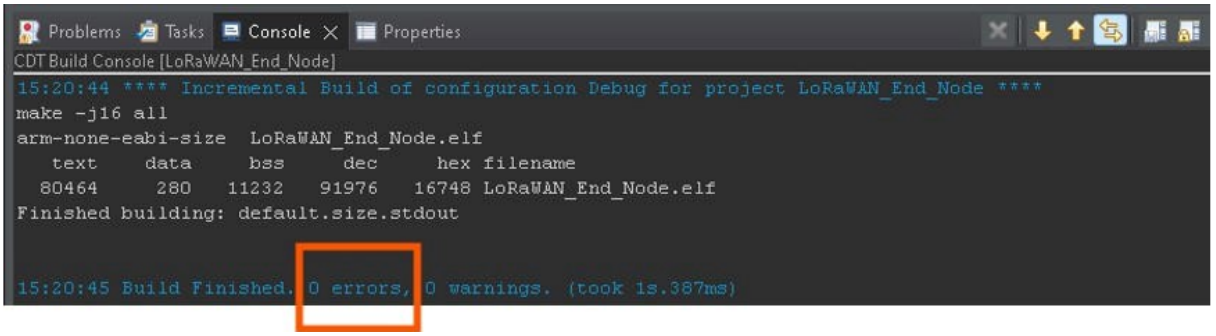

Operation completed...
Successfully imported the project 'LoRaWAN_End_Node' into the workspace.
OK


Microsoft Defender Exclusion Check
Microsoft Windows Defender is active and could significantly decrease the startup and overall performance of STM32CubeIDE.
Select how this installation should be handled by Microsoft Defender:
Exclude STM32CubeIDE from being scanned to improve performance.
(In general adding exclusions may affect the security level of this computer)
Keep STM32CubeIDE being scanned by Microsoft Defender.
See 'Startup and Shutdown' preference for more details and configuration options.
Proceed
Cancel


Exclusion failed
Cannot exclude STM32CubeIDE from being scanned by Microsoft Defender.
Tamper protection for antivirus exclusions is enabled.
OK

- Kompiliere das Projekt fehlerfrei :



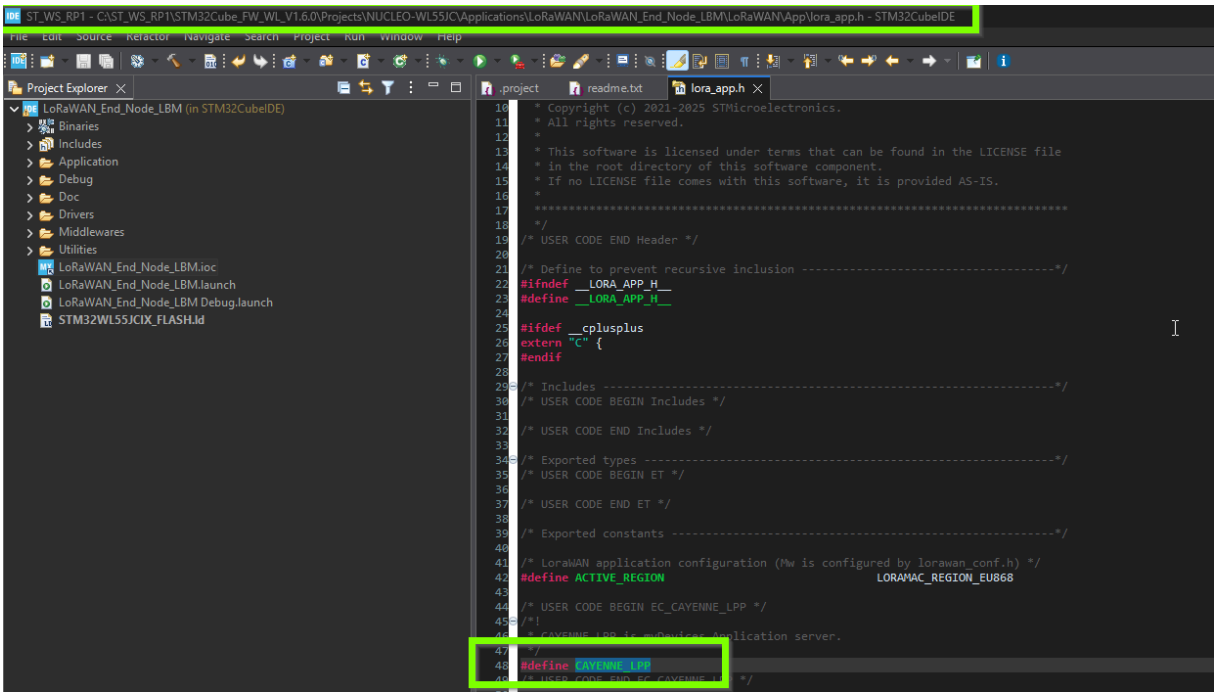


Programmiercode-Änderung

- Aktiviere “Low Power Payload” von Cayenne:
    - Entkommentiere /*#define CAYENNE_LPP in “lora_app.h”*/:
      #define CAYENNE_LPP

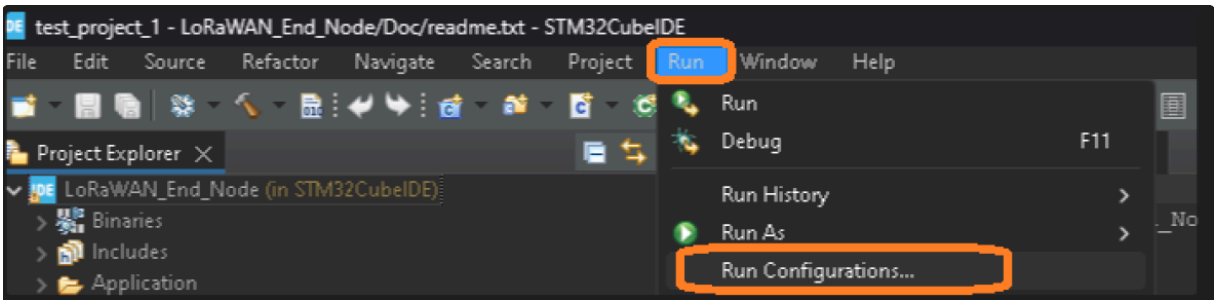
"C:\ST_WS_RP1\STM32Cube_FW_WL_V1.6.0\Projects\NUCLEO-WL55JC\Applications\LoRaWAN\LoRaWAN_End_Node_LBM\LoRaWAN\App\lora_app.h"

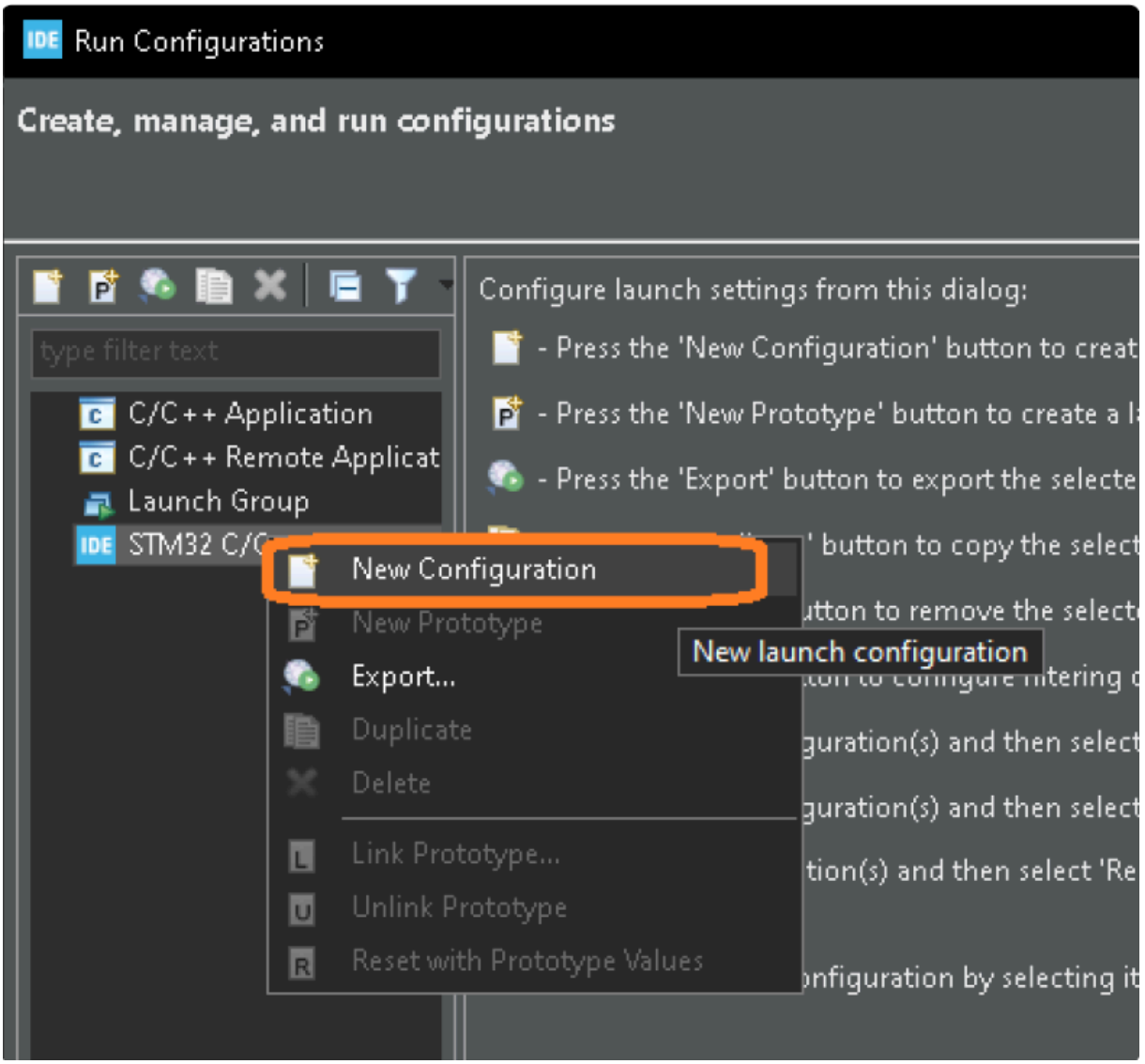
Hinweis: Cayenne LPP ist ein kompaktes Datenformat zur Kodierung von Sensordaten für LoRaWAN. Mehr Infos unter diesem Link:
Was ist “Cayenne Low Power Payload”? 🤔🤷‍♂️



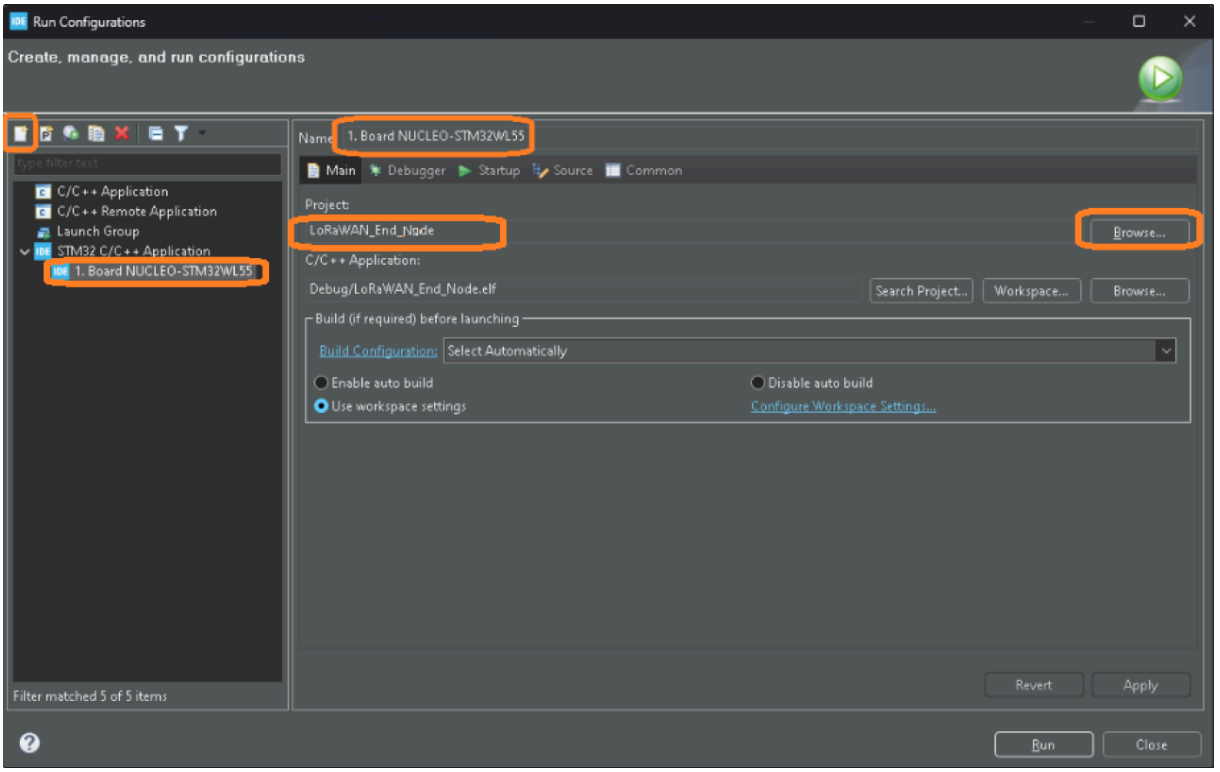
Board Einstellungen

- Erstelle für das Entwicklungsboard "new konfiguration"

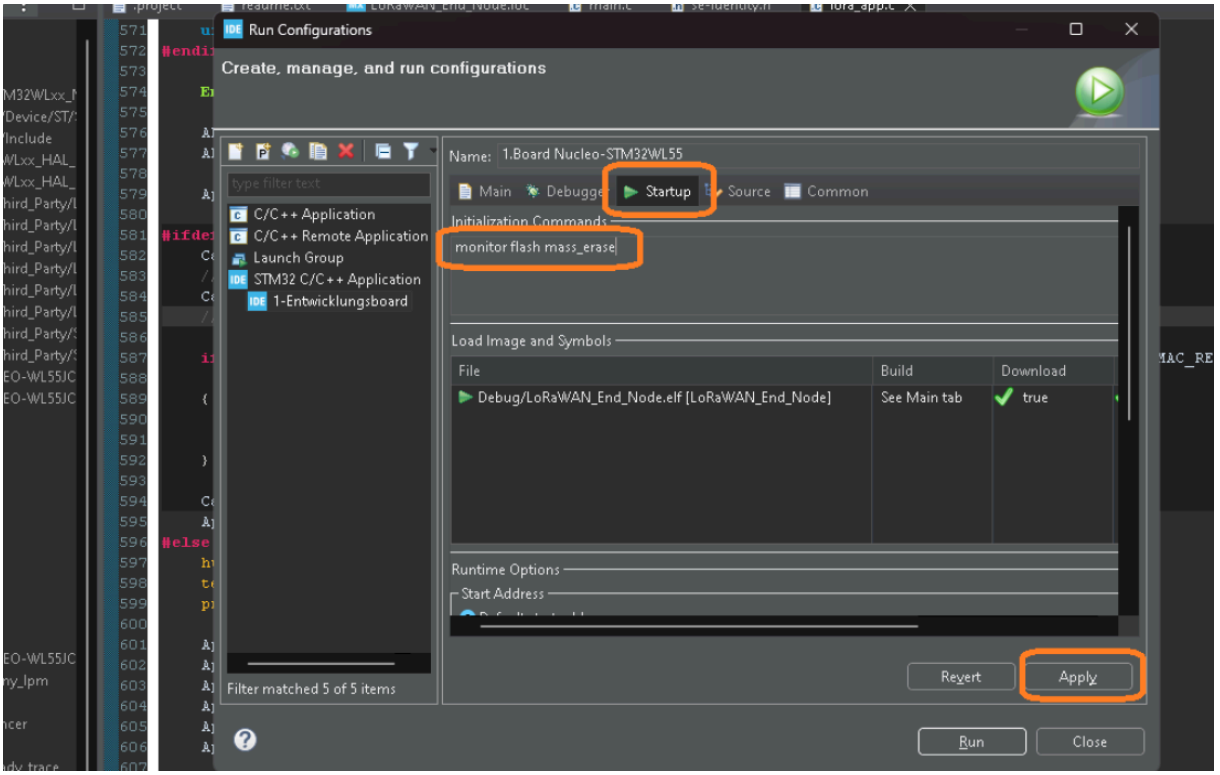




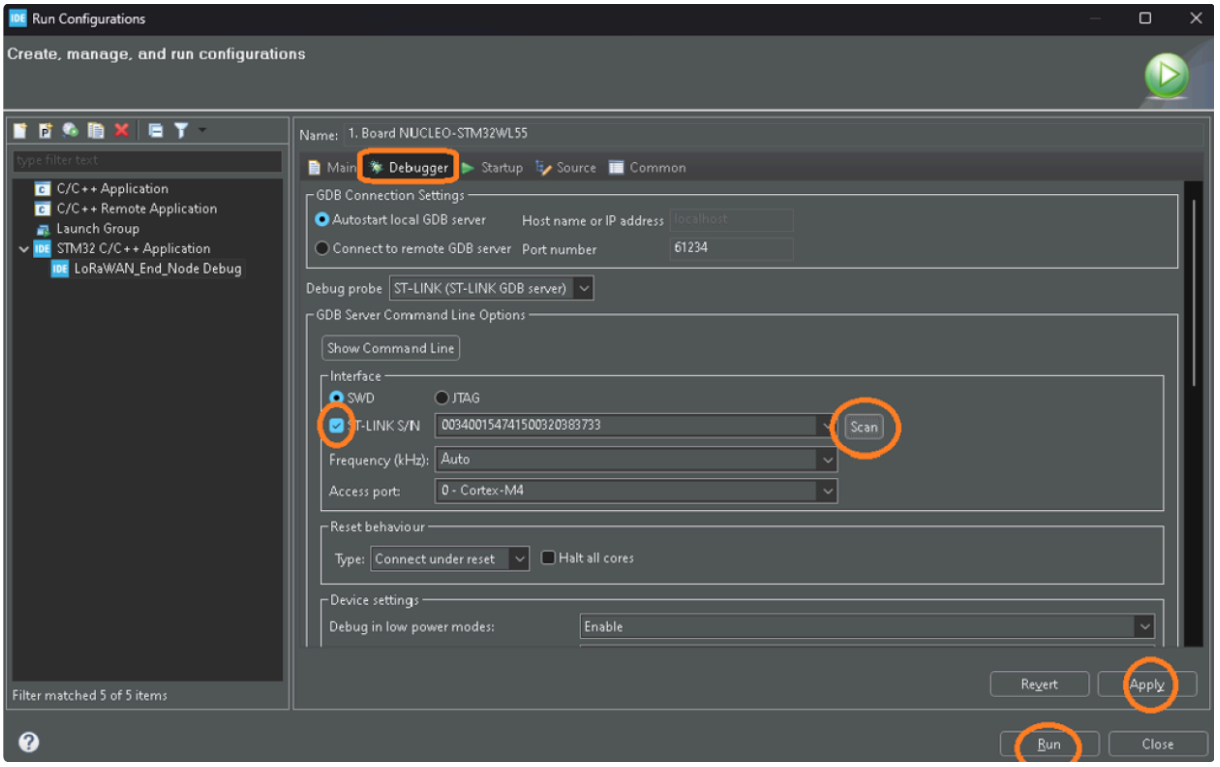
- Benenne dein Board NUCLEO-STM32WL55



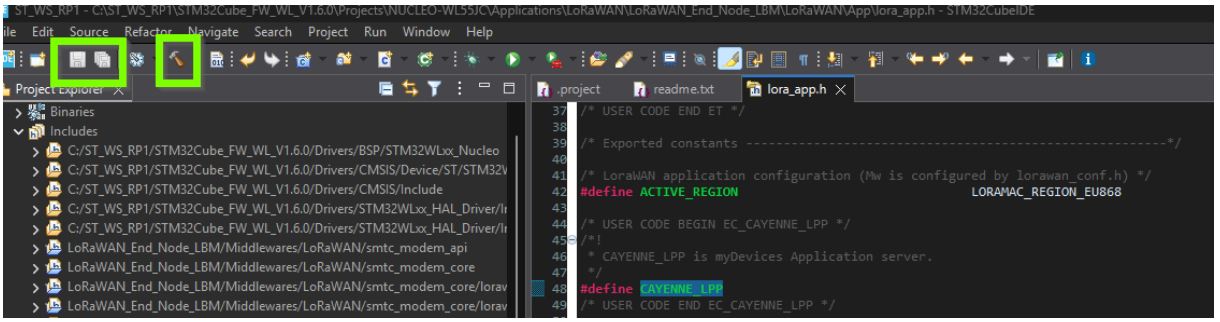
- Füge unter *Initialization Commands* folgenden Befehl hinzu:
  "*monitor flash mass_erase*"



- Entwicklungsboard per USB mit dem PC verbinden

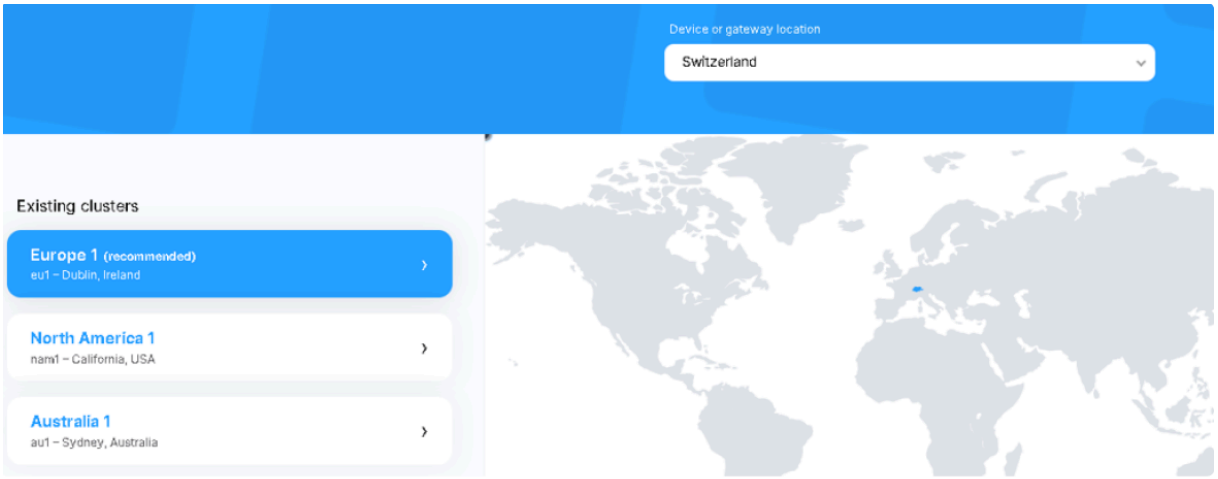


- Speichere die Konfiguration



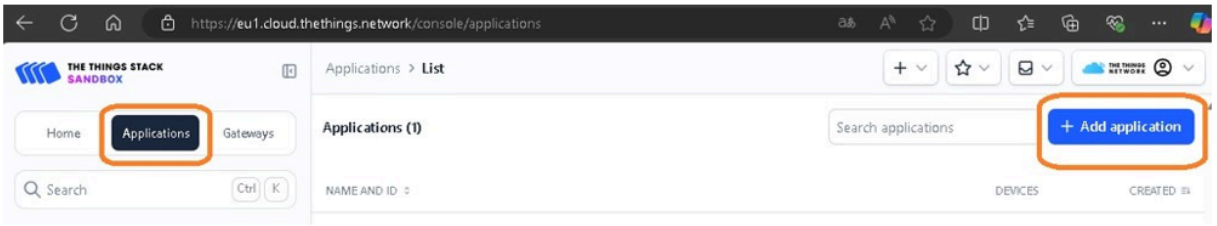
“The Thing Network” (TTN) / “The Stacks”

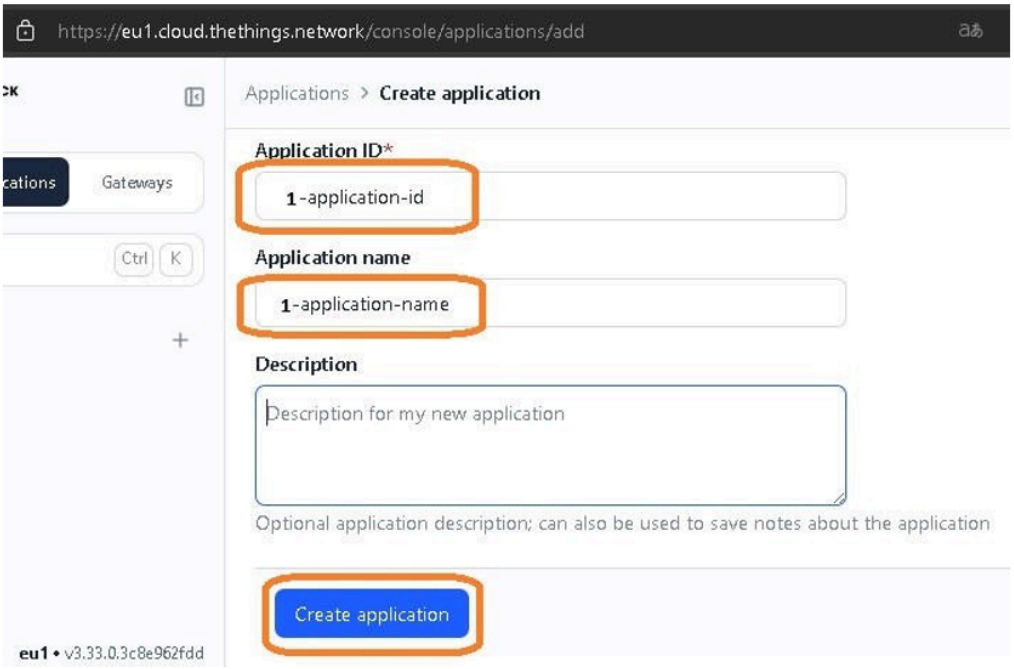
- Erstelle eine Konto, um Zugriff auf “The things network” zu erhalten.
- Wähle Cluster Europe 1(Für Schweiz)

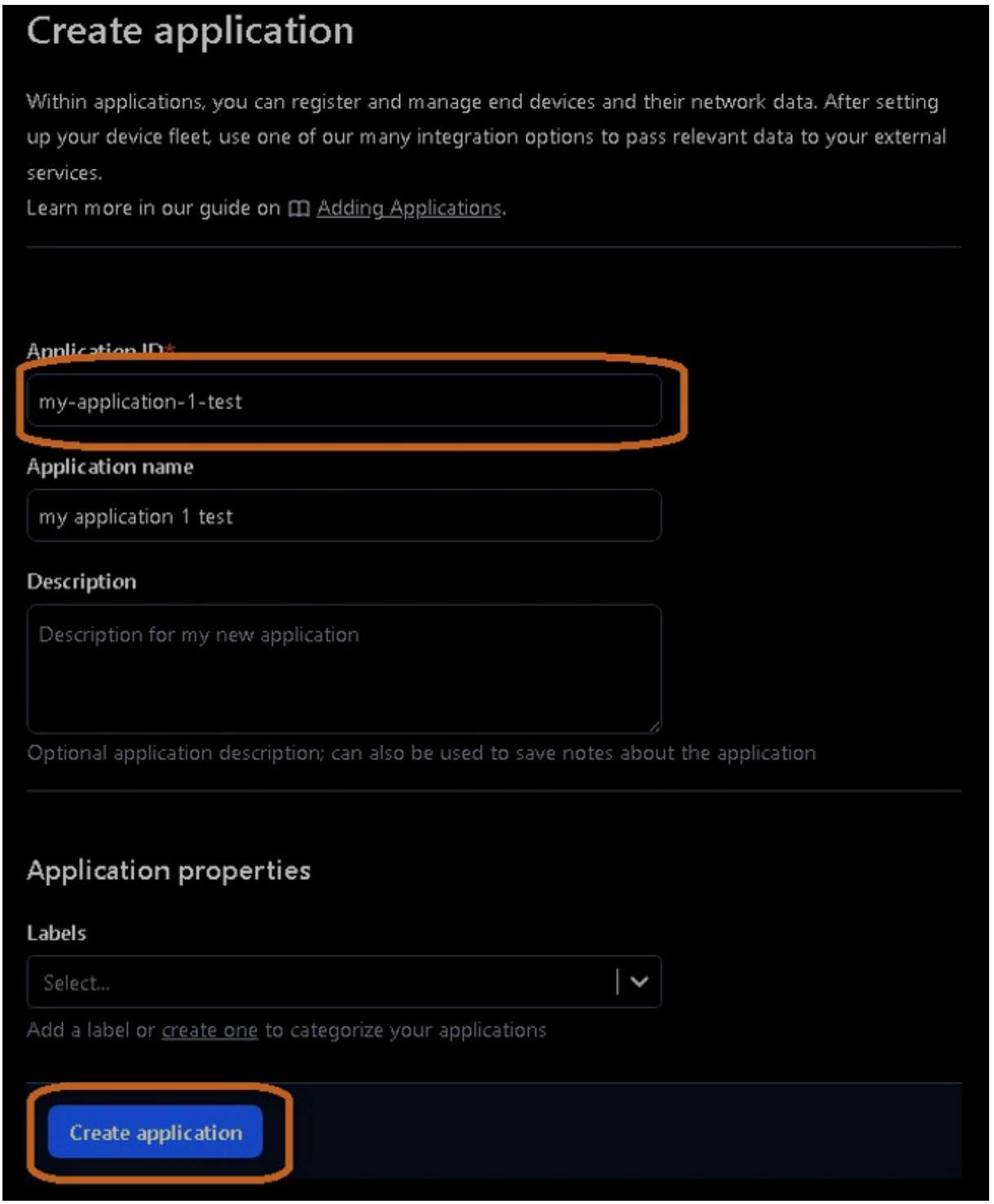


Cluster Für die Schweiz: Europe 1

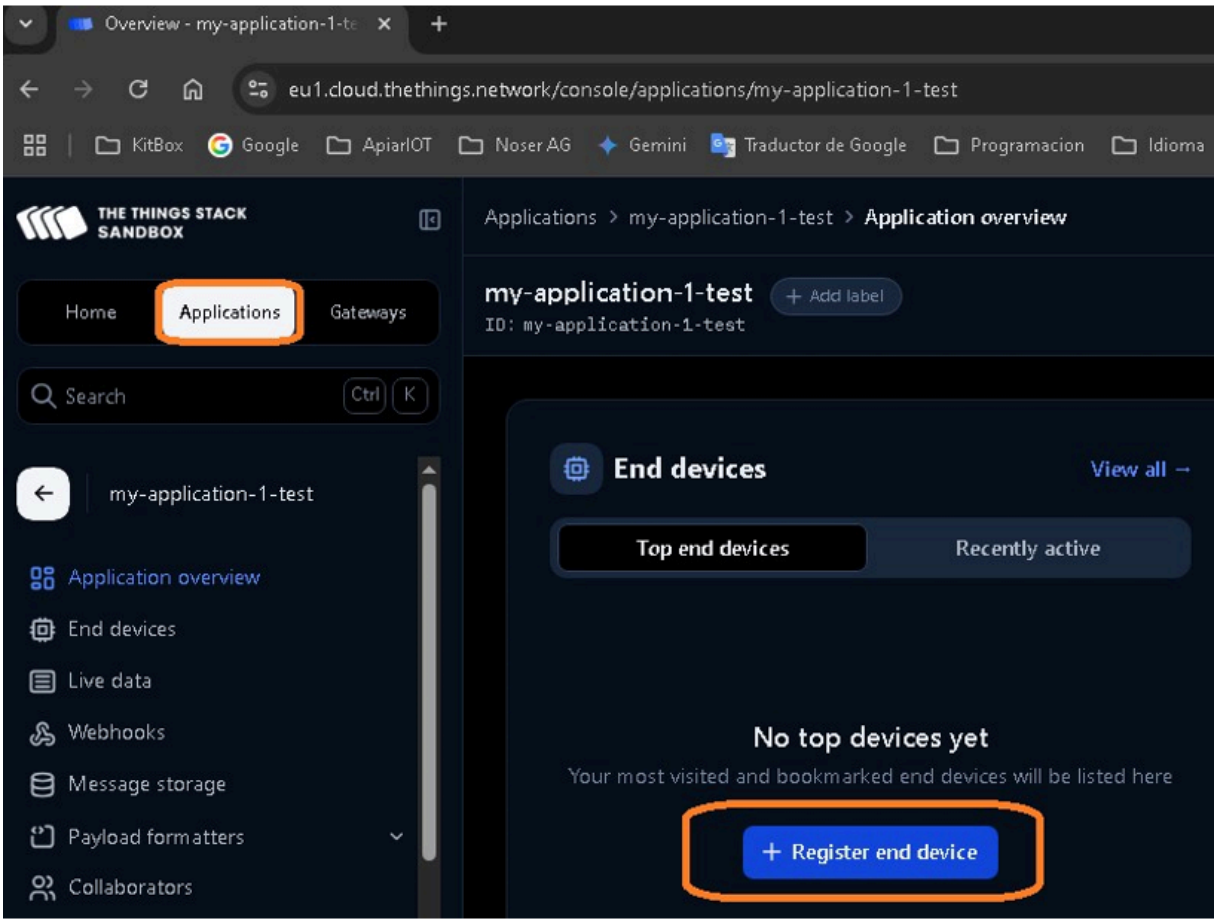
- Erstelle eine Applikation



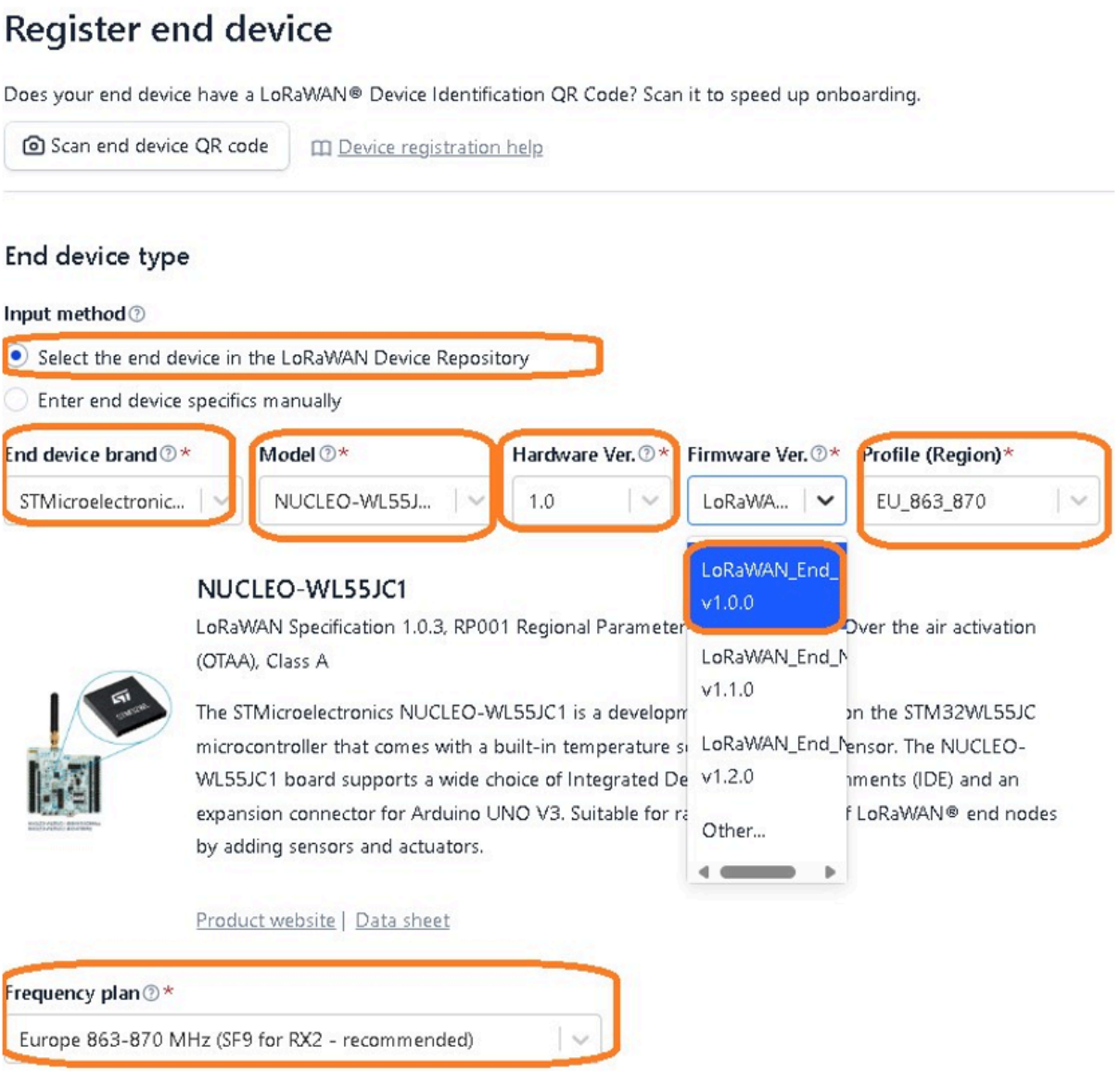



Create application
Within applications, you can register and manage end devices and their network data. After setting up your device fleet, use one of our many integration options to pass relevant data to your external services.
Learn more in our guide on Adding Applications.
Application ID*
my-application-1-test
Application name
my application 1 test
Description
Description for my new application
Optional application description; can also be used to save notes about the application
Application properties
Labels
Select...
Add a label or create one to categorize your applications
Create application

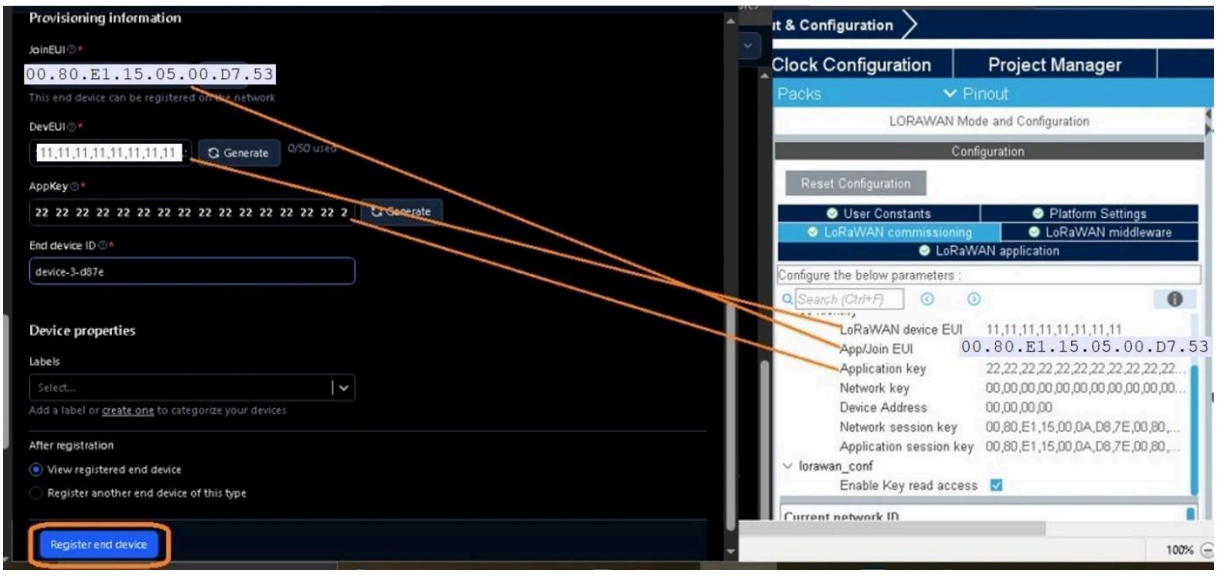
- Registriere das "End-device".

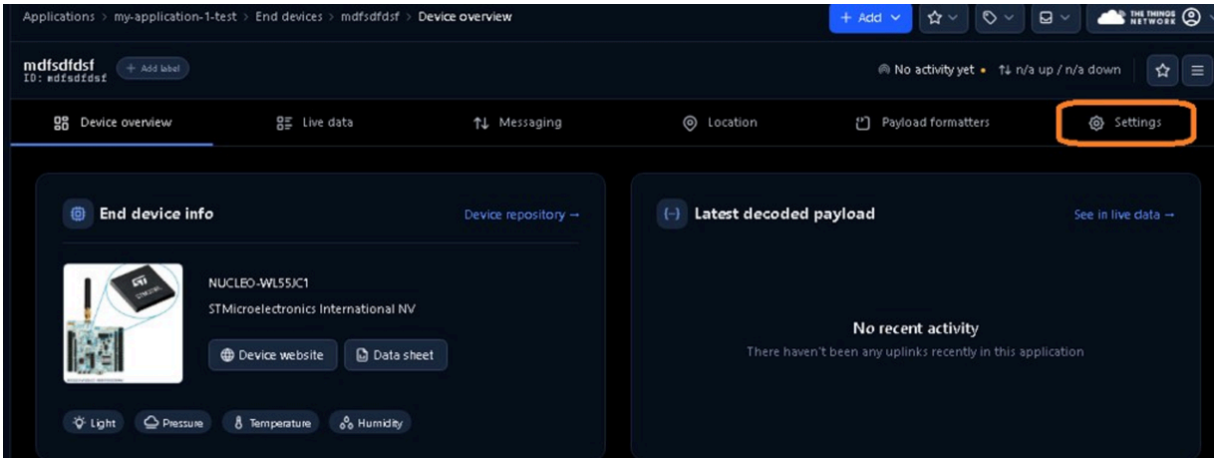


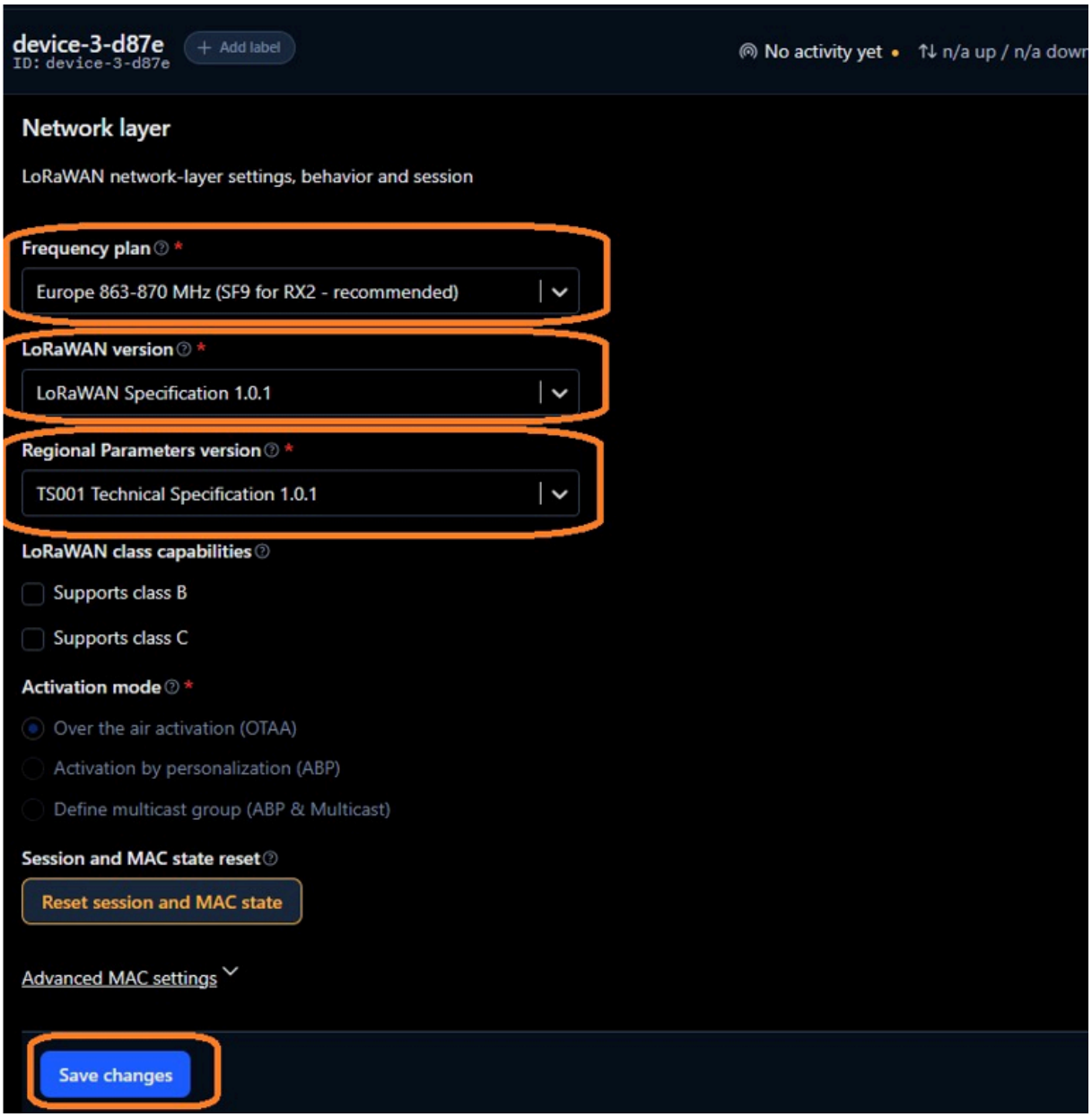
- End device type, End device brand, Model, Hardware version, Firmware version, Profile region, Frequency plan



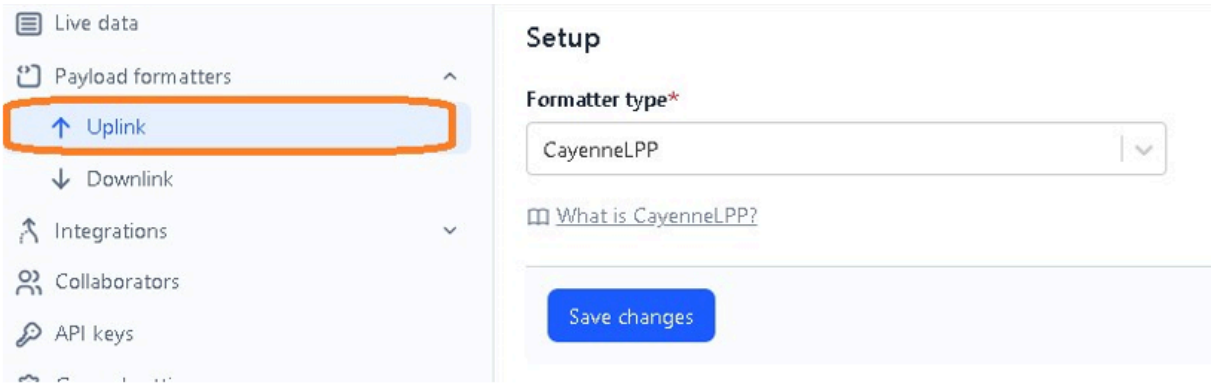
- Suche die Information von "LoRaWAN commissioninig"

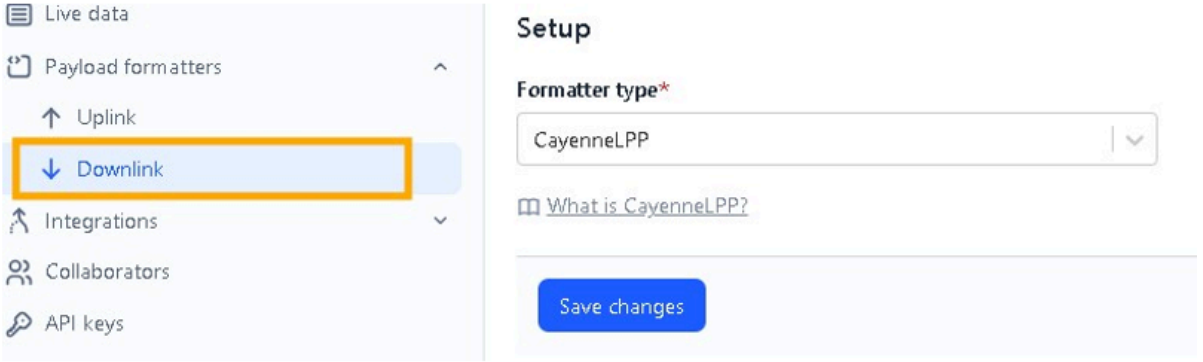



Applications > my-application-1-test > End devices > mdfsdfdsf > Device overview
+ Add
mdfsdfdsf
ID: mdfsdfdsf
+ Add label
No activity yet • ↑↓ n/a up / n/a down
Device overview
Live data
Messaging
Location
Payload formatters
Settings
End device info
Device repository →
NUCLEO-WL55JC1
STMicroelectronics International NV
Device website
Data sheet
Light
Pressure
Temperature
Humidity
Latest decoded payload
See in live data →
No recent activity
There haven't been any uplinks recently in this application

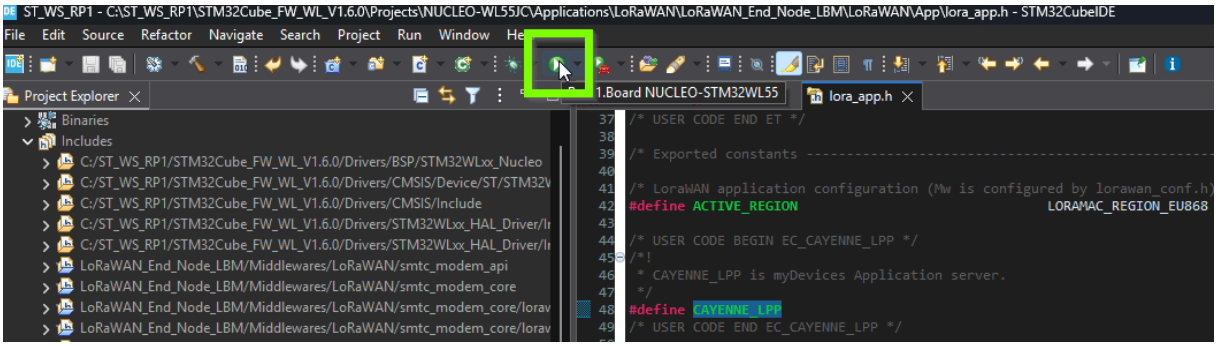

device-3-d87e
ID: device-3-d87e
+ Add label
No activity yet • ↑↓ n/a up / n/a down
Network layer
LoRaWAN network-layer settings, behavior and session
Frequency plan *
Europe 863-870 MHz (SF9 for RX2 - recommended)
LoRaWAN version *
LoRaWAN Specification 1.0.1
Regional Parameters version *
TS001 Technical Specification 1.0.1
LoRaWAN class capabilities
Supports class B
Supports class C
Activation mode *
Over the air activation (OTAA)
Activation by personalization (ABP)
Define multicast group (ABP & Multicast)
Session and MAC state reset
Reset session and MAC state
Advanced MAC settings
Save changes

- Konfiguriere den "Uplink" und den "Downlink" für die “Payload” von “End-device

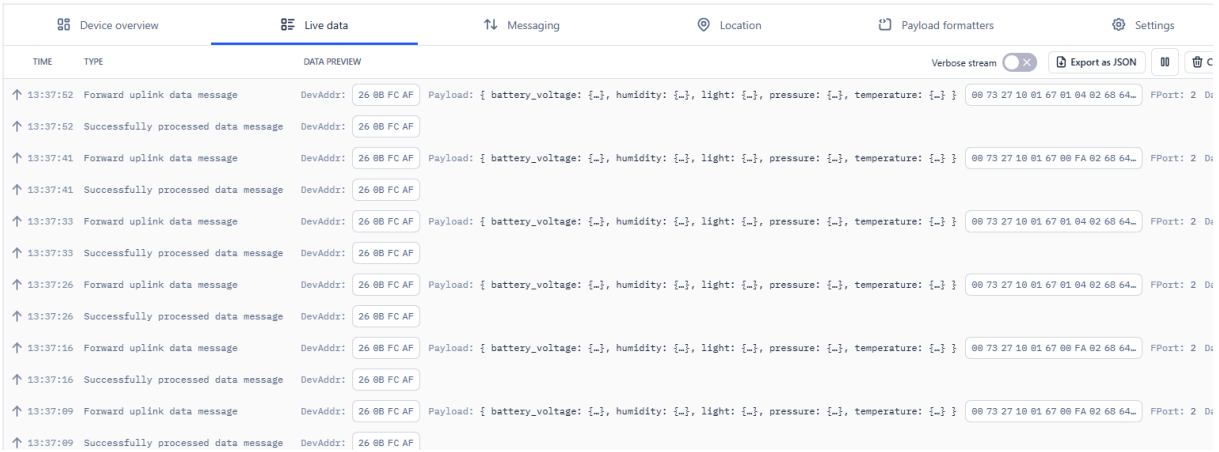




- Lade das Programm auf das Board

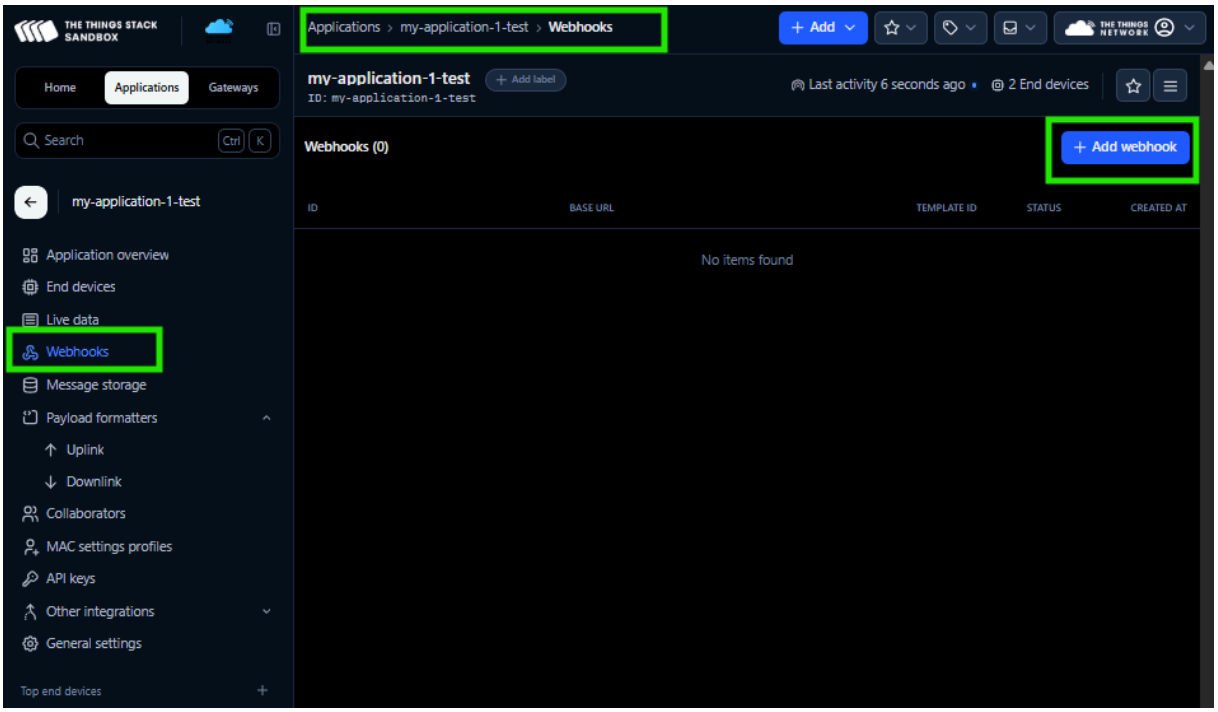


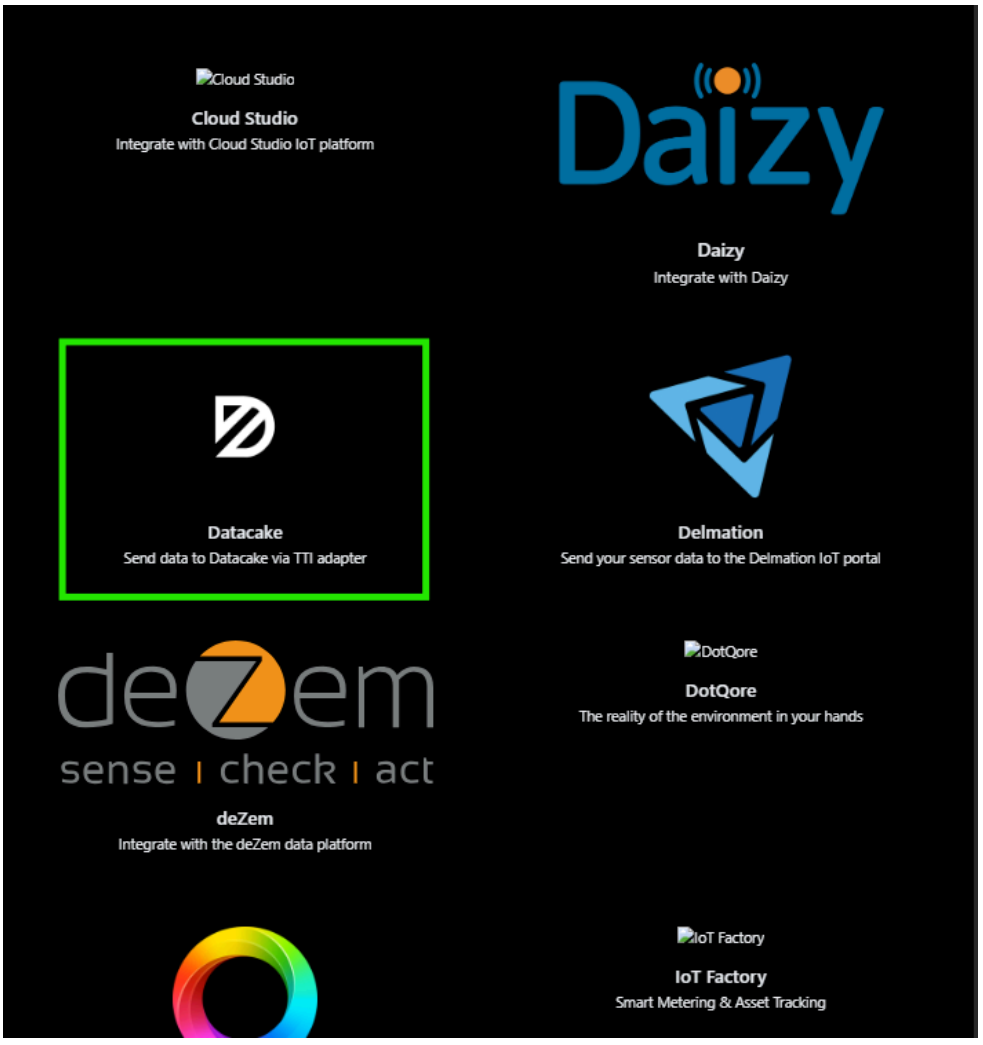
- Prüfe, ob TTN die Daten bekommt, wenn ja schalte dein Entwicklungsboard aus.

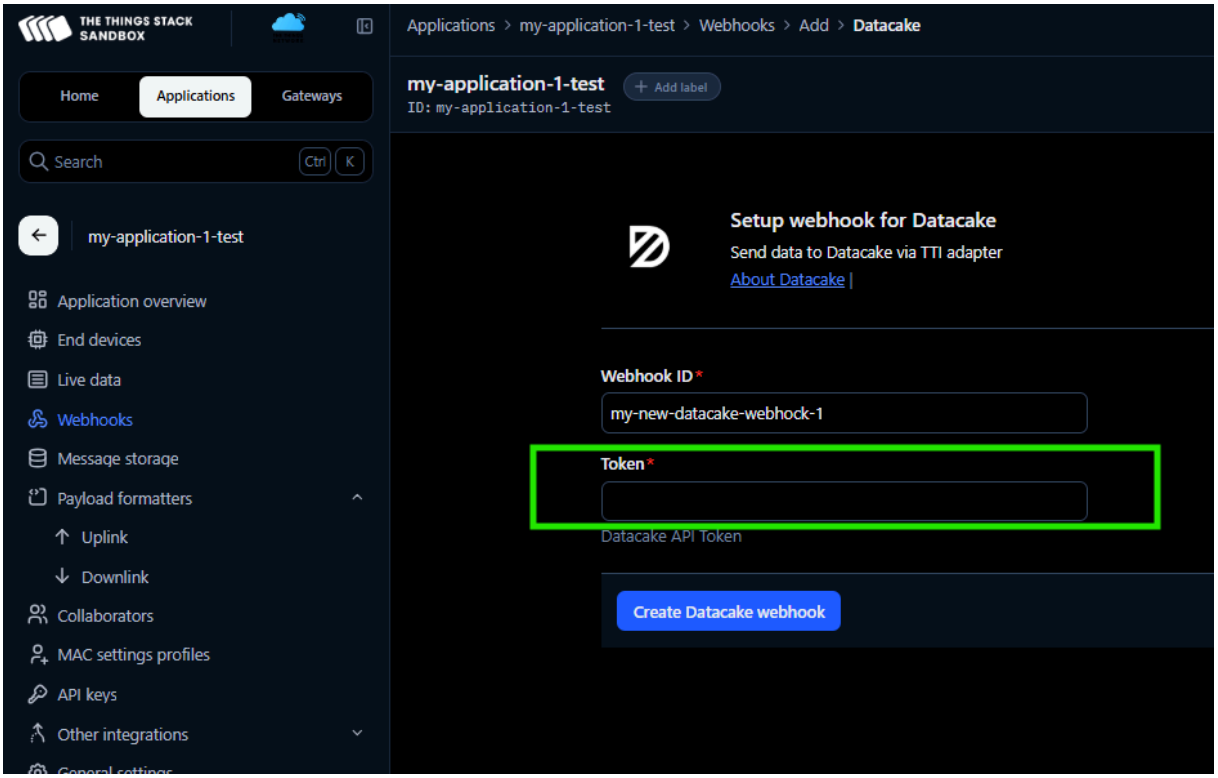


Uplink forwarding TTN to Datacake

- Webhooks einrichten

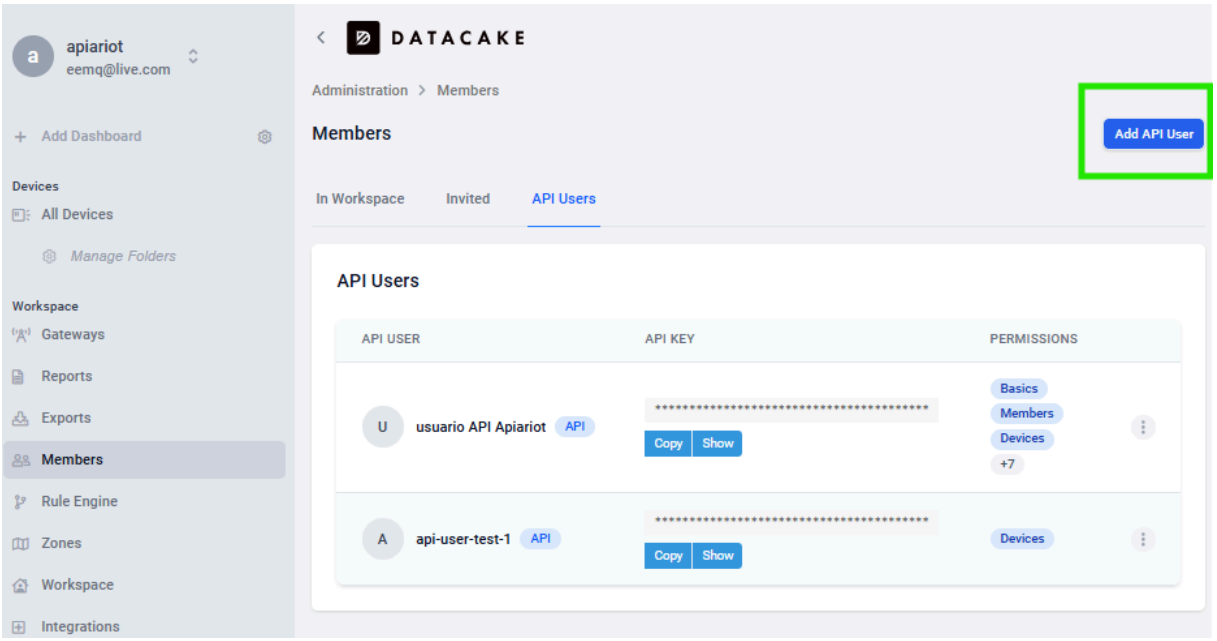


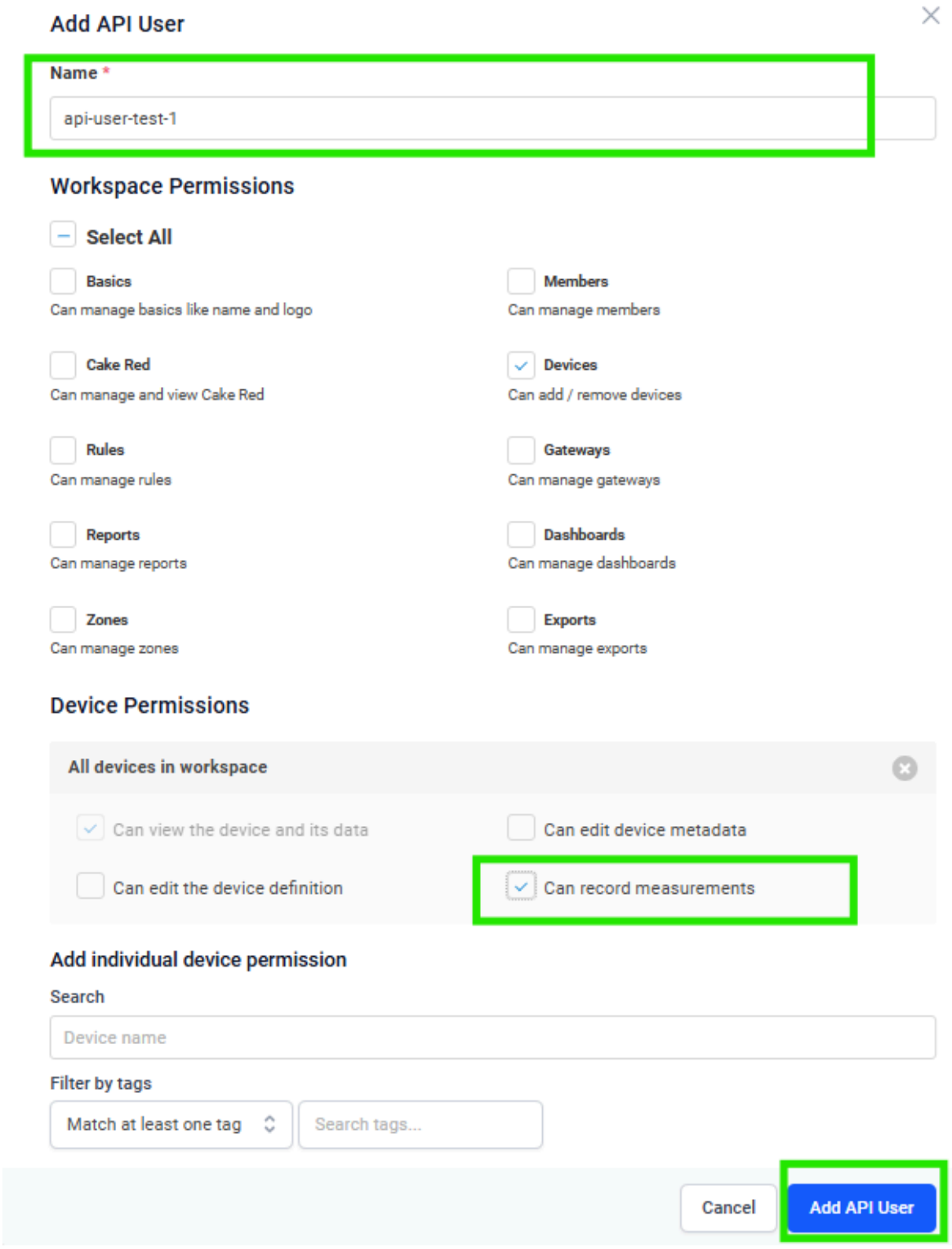




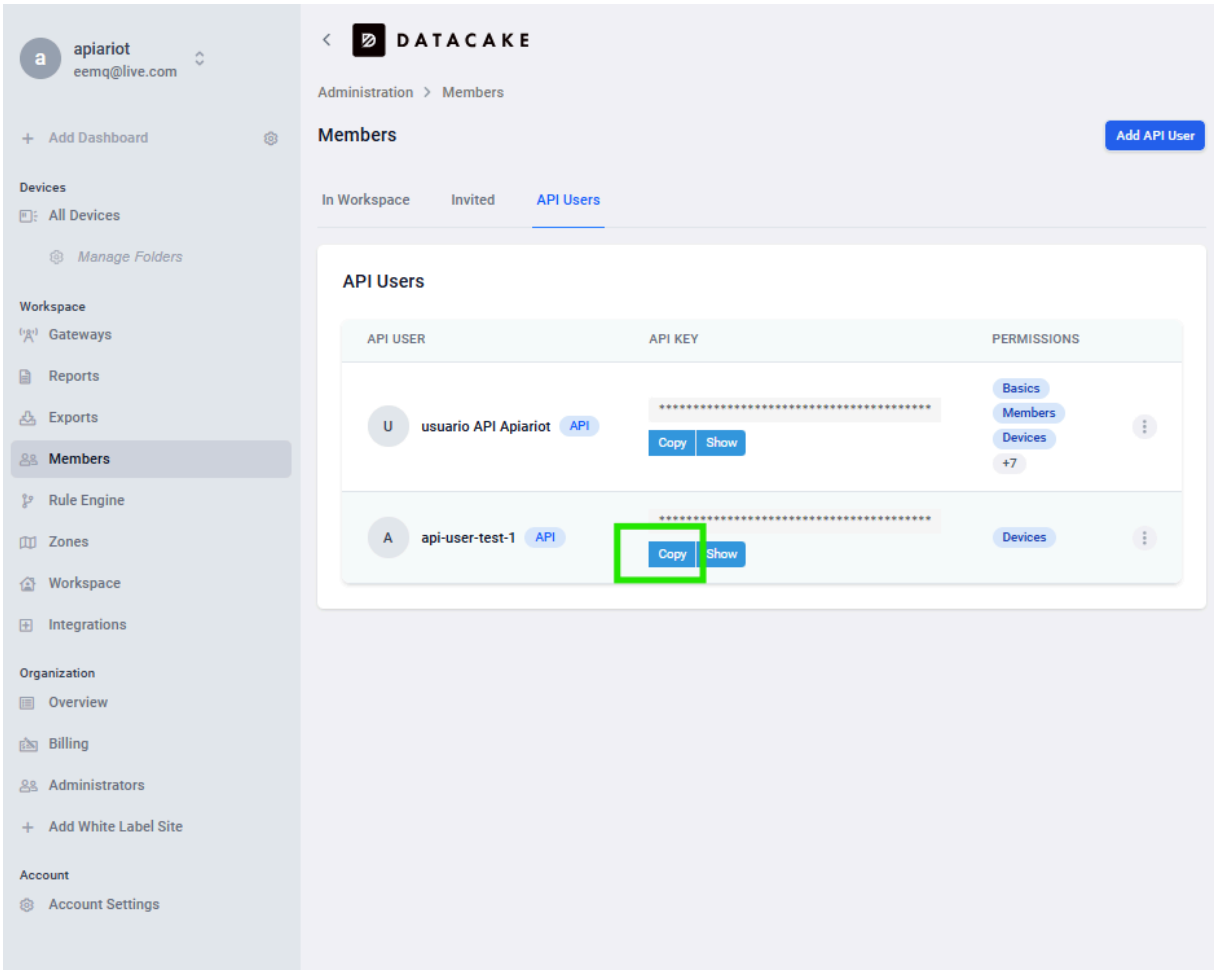
- Du benötigst für den Token zuerst ein Konto bei Datacake.
- Erstelle dein Konto in [Datacake.]

- Und füge einen "API User" hinzu

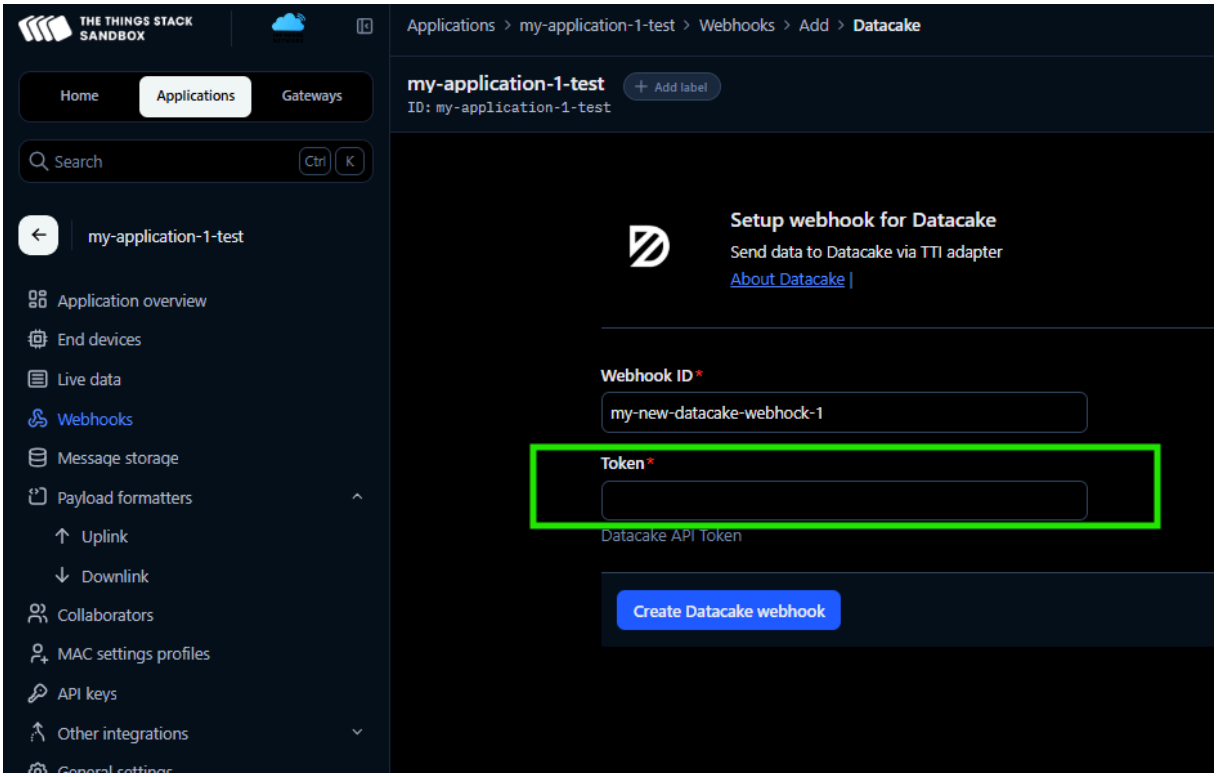



Add API User
Name *
api-user-test-1
Workspace Permissions
Select All
Basics
Can manage basics like name and logo
Members
Can manage members
Cake Red
Can manage and view Cake Red
Devices
Can add / remove devices
Rules
Can manage rules
Gateways
Can manage gateways
Reports
Can manage reports
Dashboards
Can manage dashboards
Zones
Can manage zones
Exports
Can manage exports
Device Permissions
All devices in workspace
Can view the device and its data
Can edit device metadata
Can edit the device definition
Can record measurements
Add individual device permission
Search
Device name
Filter by tags
Match at least one tag
Search tags...
Cancel
Add API User

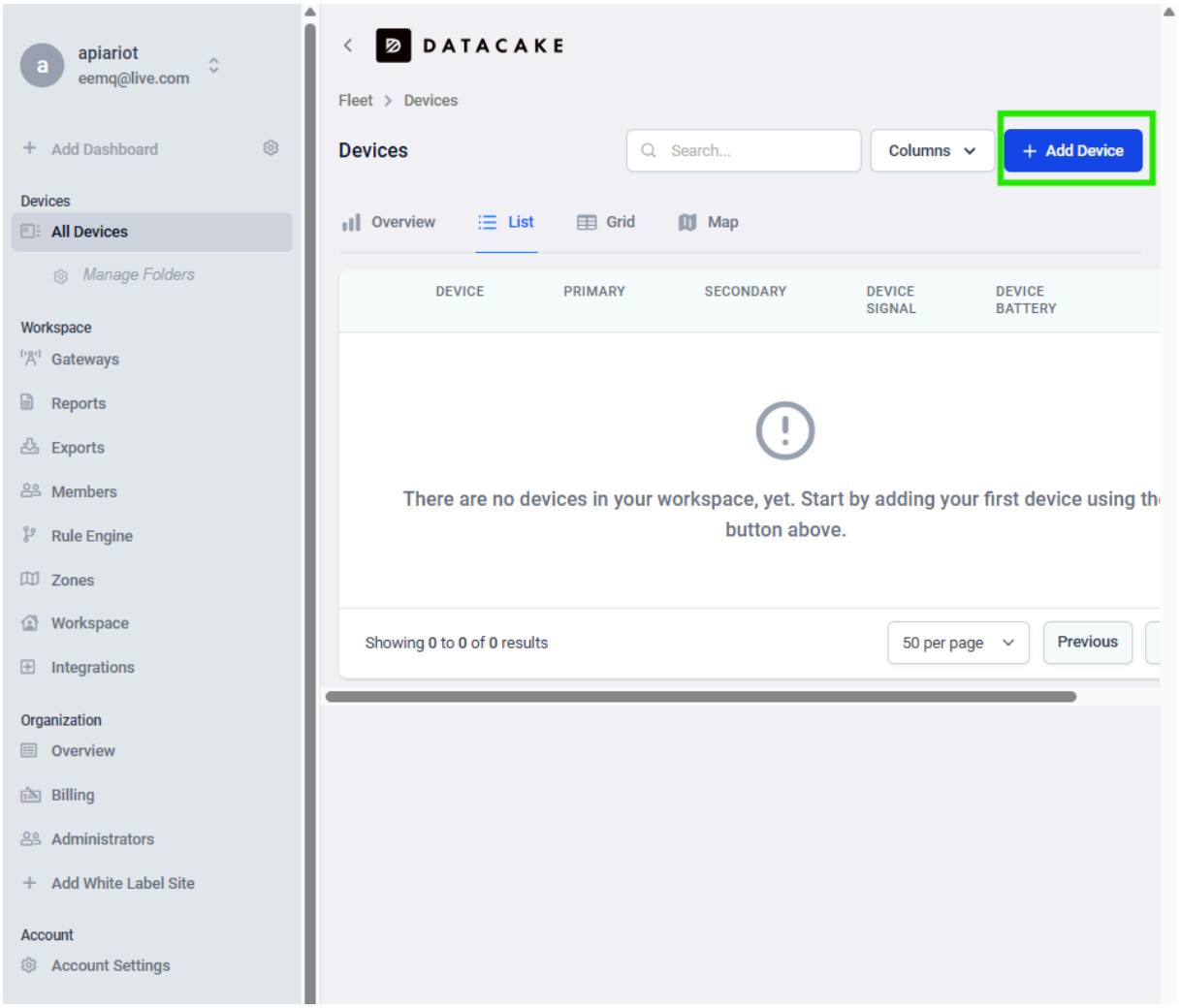
- Kopiere das Token für TTN

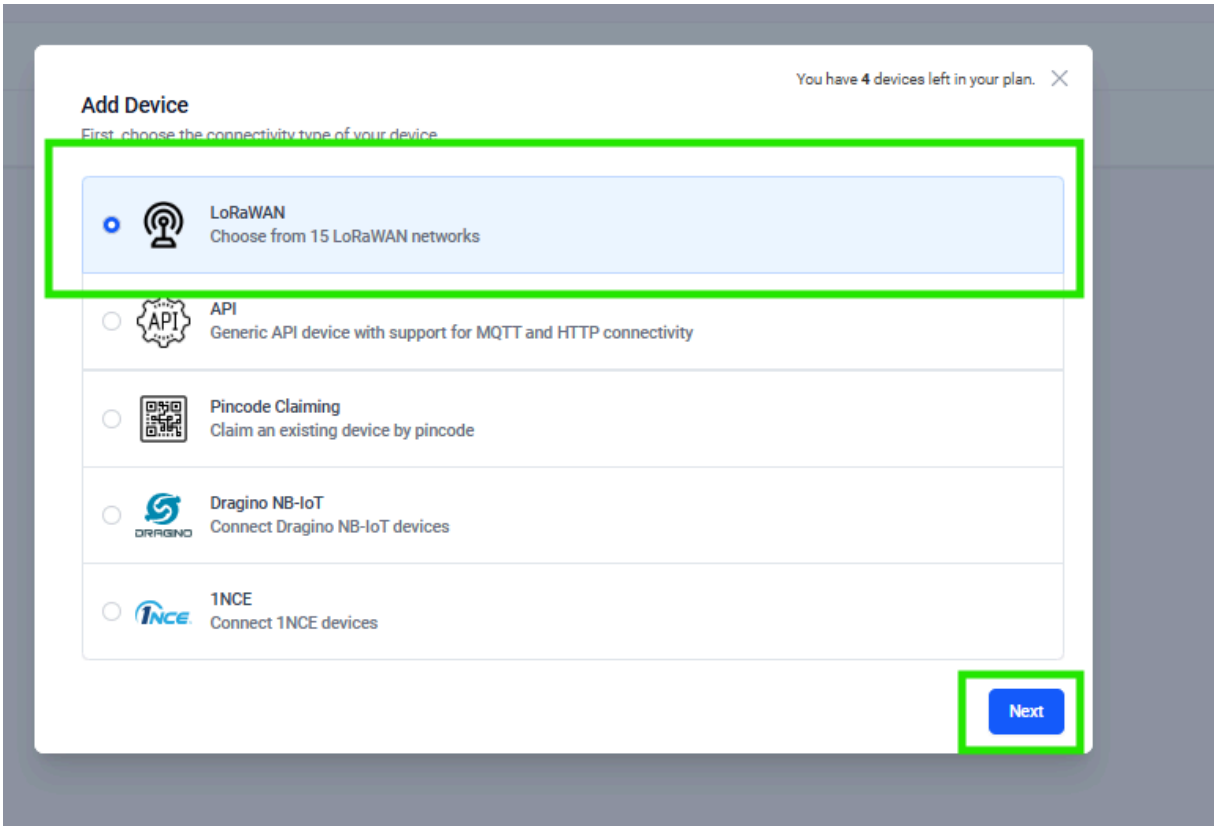


- Du hast jetzt den Token für TTN

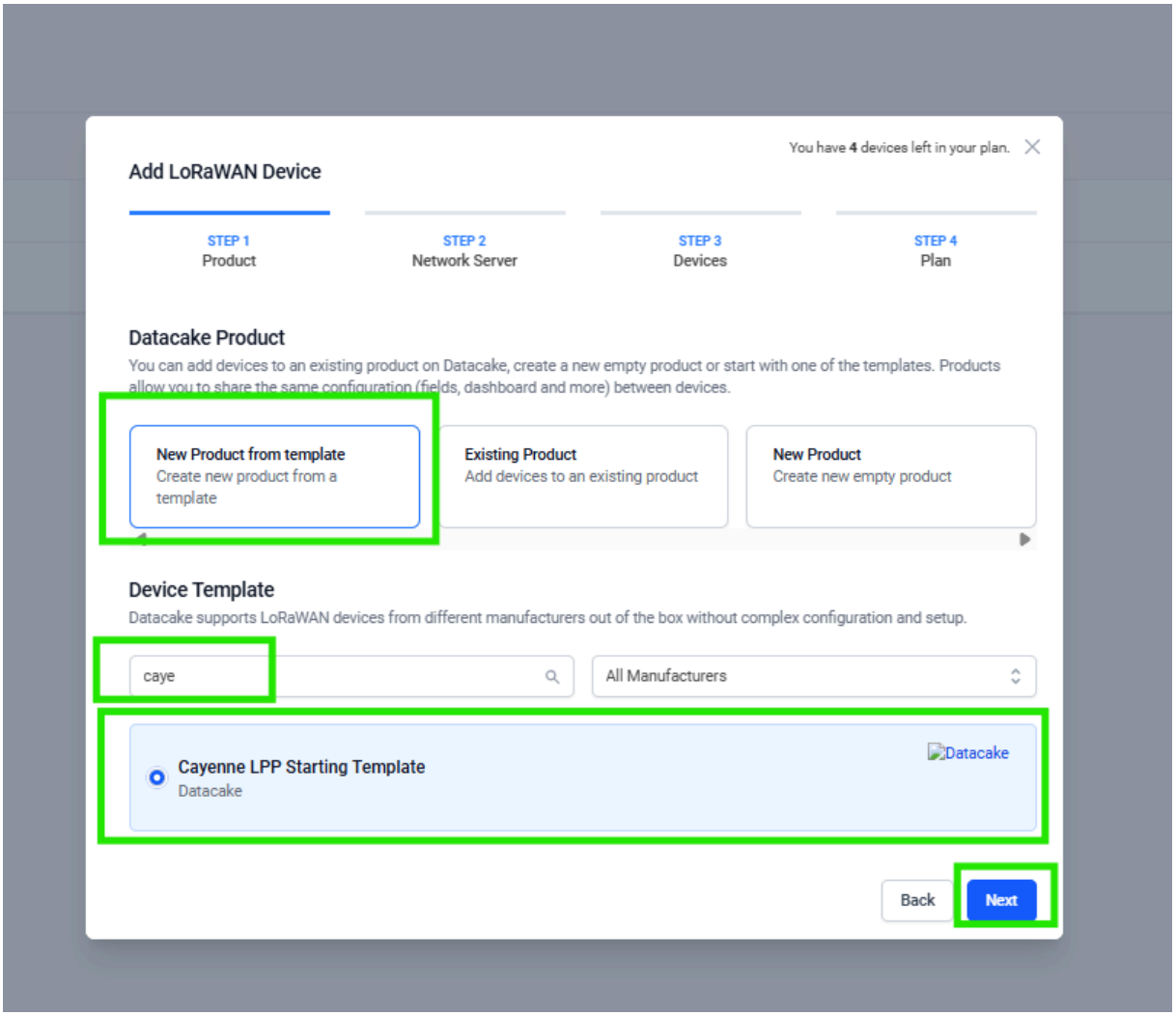


- Jetzt bei Datacake, das Gerät hinzufügen




You have 4 devices left in your plan.
Add Device
First, choose the connectivity type of your device
LoRaWAN
Choose from 15 LoRaWAN networks
API
Generic API device with support for MQTT and HTTP connectivity
Pincode Claiming
Claim an existing device by pincode
Dragino NB-IoT
Connect Dragino NB-IoT devices
1NCE
Connect 1NCE devices
Next

- Verwende als Netzwer Server: eu1.cloud.thethings.network (Schweiz)



SCHRITT 1
Produkt

SCHRITT 2
Netzwerk-Server

SCHRITT 3
Geräte

SCHRITT 4
Plan

Netzwerk-Server

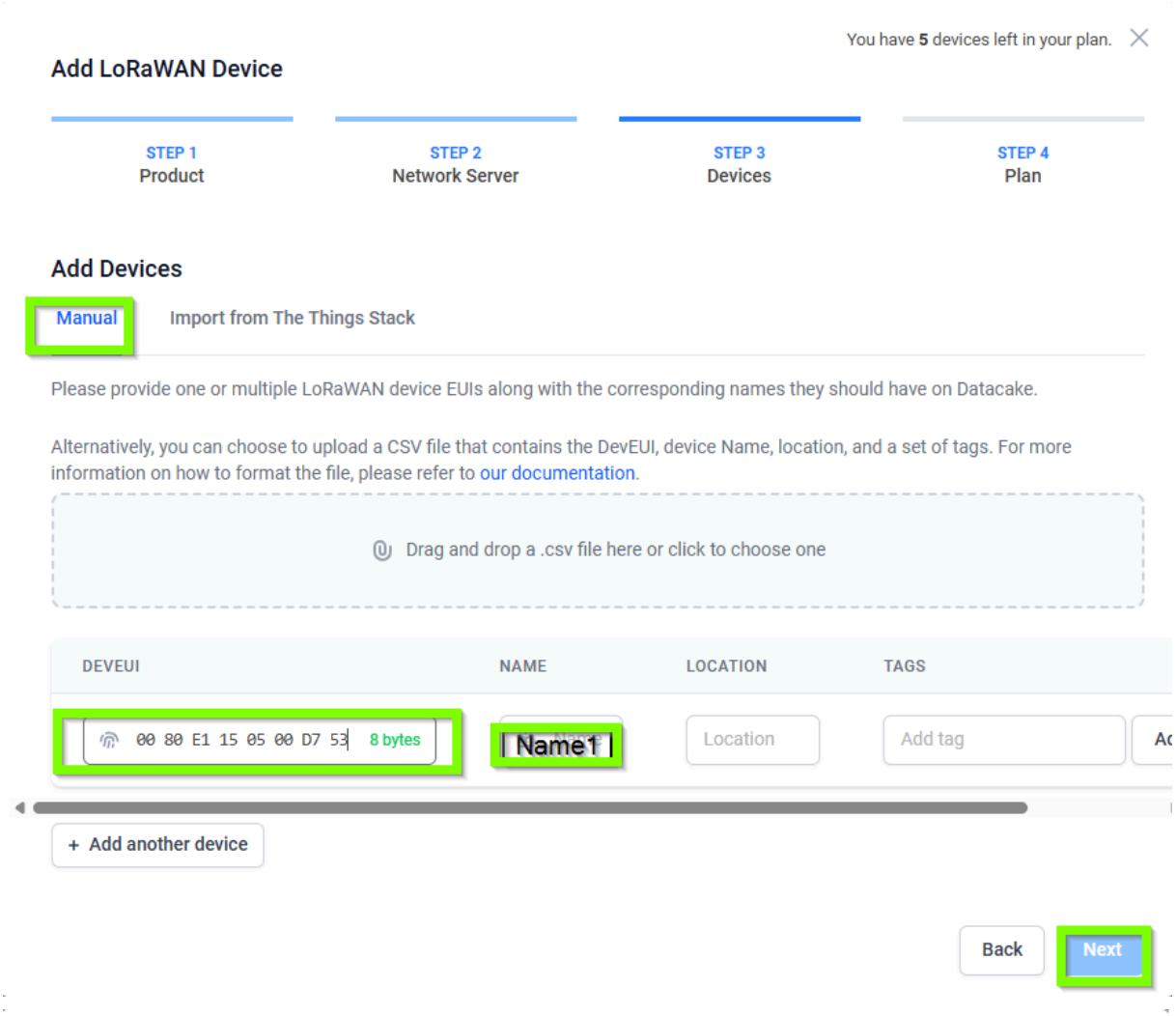
Bitte wählen Sie den LoRaWAN Network Server, mit dem Ihre Geräte verbunden sind.

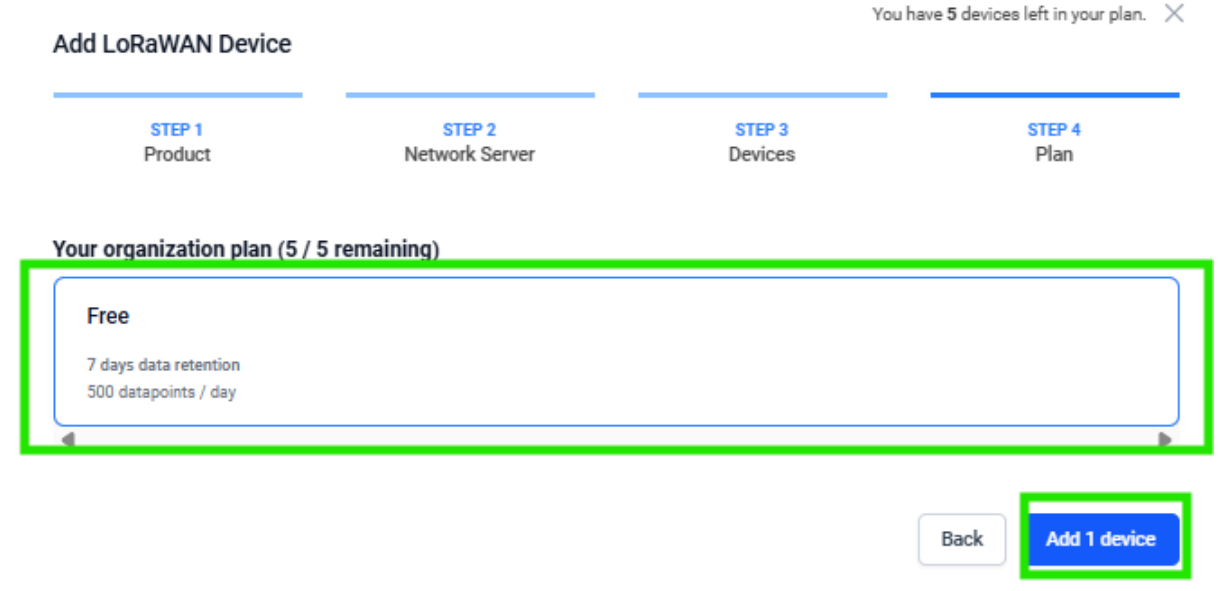
○	**Datacake LNS** AUTOMATIC SETUP Einfach starten und skalieren mit einem verwalteten LNS	Uplinks Downlinks
◉	**The Things Stack V3** TTN V3 / Things Industries	Uplinks Downlinks
○	**ThingPark**	Uplinks Downlinks
○	**KPN**	Uplinks Downlinks
○	**LORIOT**	Uplinks Downlinks
○	**ChirpStack**	Uplinks Downlinks
○	**Helium** Verwenden Sie Ihre eigene Konsole	Uplinks Downlinks

Anzeigen von **1** bis **6** von **14** Ergebnissen

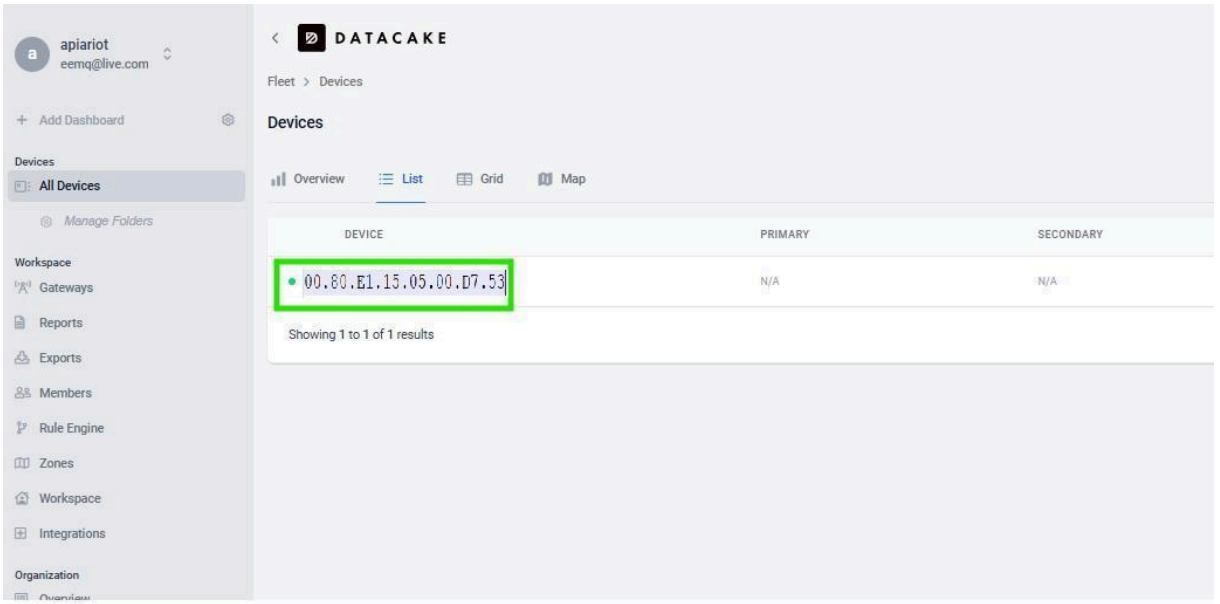
Vorherige Weiter

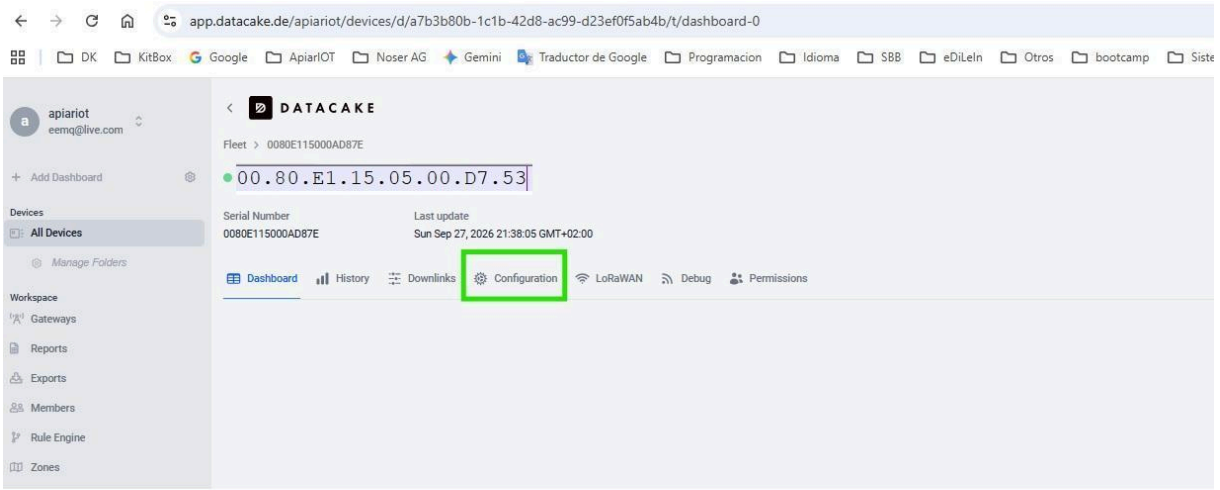
Zurück Weiter


You have 5 devices left in your plan.
Add LoRaWAN Device
STEP 1
Product
STEP 2
Network Server
STEP 3
Devices
STEP 4
Plan
Add Devices
Manual
Import from The Things Stack
Please provide one or multiple LoRaWAN device EUIs along with the corresponding names they should have on Datacake.
Alternatively, you can choose to upload a CSV file that contains the DevEUI, device Name, location, and a set of tags. For more information on how to format the file, please refer to our documentation.
Drag and drop a .csv file here or click to choose one
DEVEUI
NAME
LOCATION
TAGS
00 80 E1 15 05 00 D7 53
8 bytes
Name1
Location
Add tag
+ Add another device
Back
Next

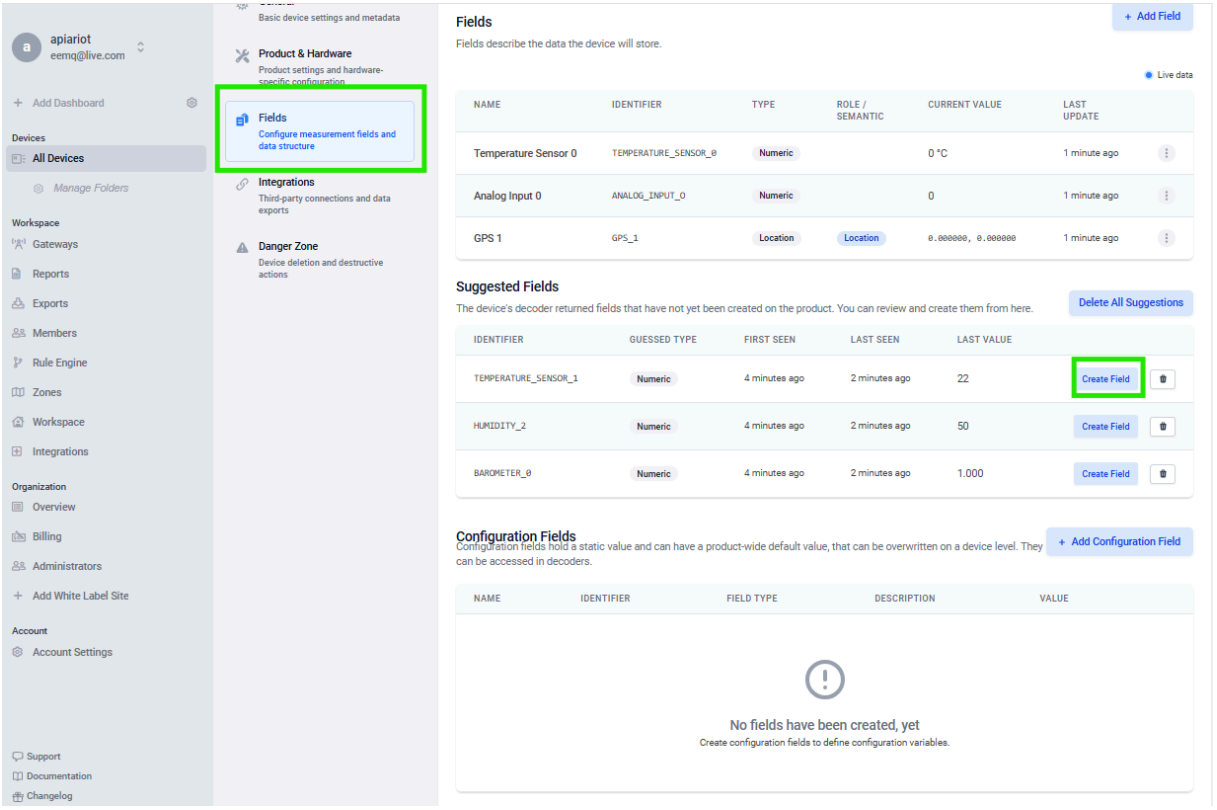

You have 5 devices left in your plan.
Add LoRaWAN Device
STEP 1
Product
STEP 2
Network Server
STEP 3
Devices
STEP 4
Plan
Your organization plan (5 / 5 remaining)
Free
7 days data retention
500 datapoints / day
Back
Add 1 device

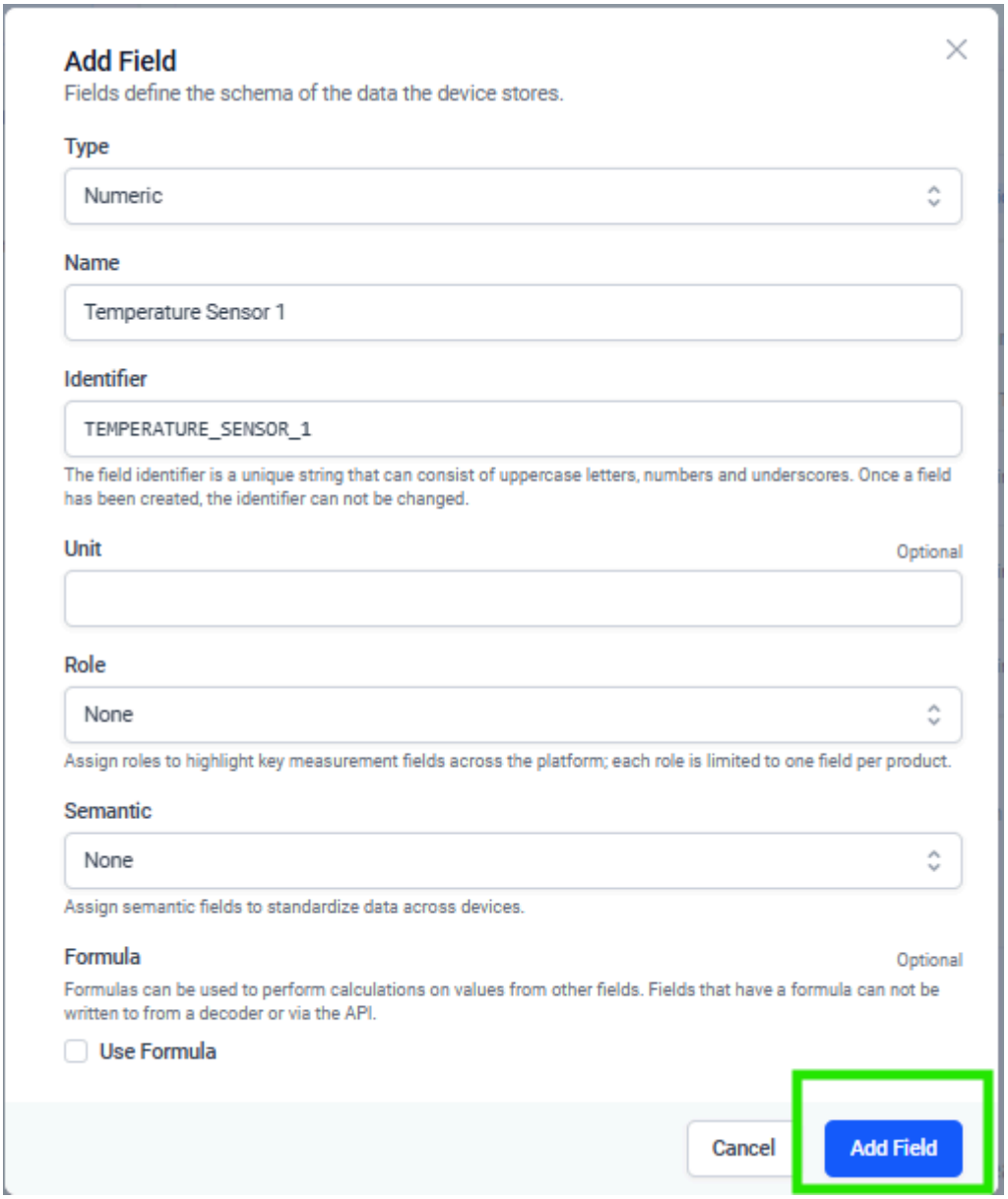
- Schalte deine "End Node" wieder ein.



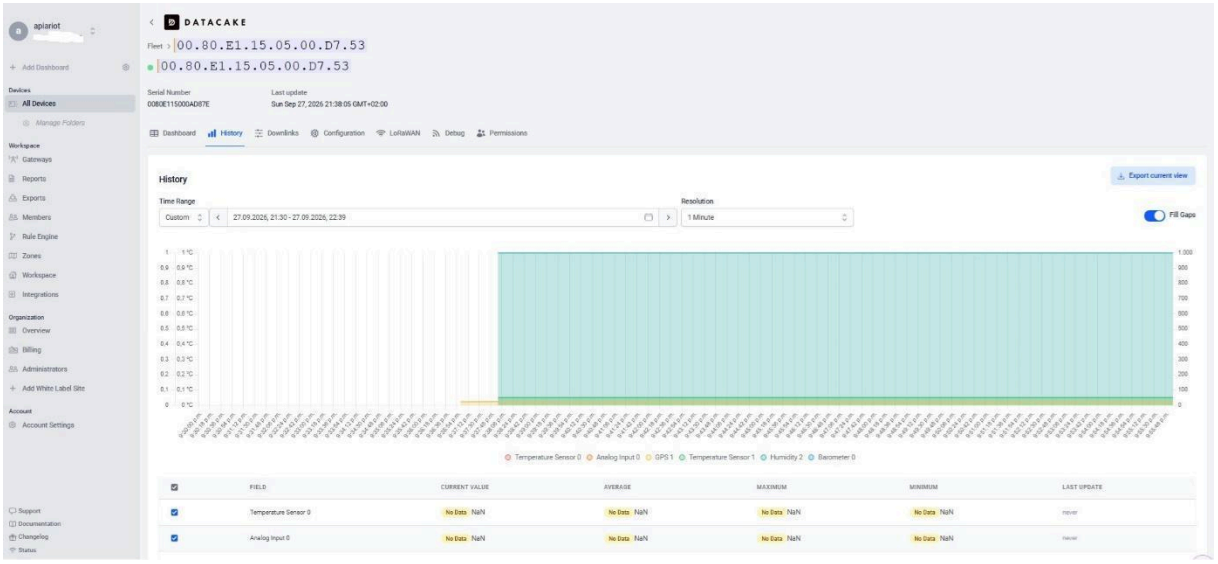


- Warte, bis Daten empfangen werden.





- Du solltest jetzt in Datacake die Daten von deinem Entwicklungsboard sehen können.



- Und in TTN ebenfalls

